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Physics-Informed Machine Learning in Design and Manufacturing: Status and Challenges

Machine learning (ML) technique is a critical tool to promote optimal design and ensure reliable and efficient products and processes in the manufacturing industry, since it can discover hidden knowledge and build complex relationships by learning patterns from data. However, the inherent 'black-box' nature of ML presents a major challenge in interpreting the mechanism and outcomes of the models. Moreover, reliable ML predictions are highly dependent on the amount and quality of training data. To address these issues, physics-informed machine learning (PIML), also known as scientific machine learning, has emerged as a new research field. PIML incorporates physical and domain knowledge into ML models to guide the ML training process, which enables more interpretable and reliable models. To fully leverage the advantages of PIML and promote the advancement of design and manufacturing, it is essential for researchers to understand the available PIML methodologies and the technical challenges of PIML methods. This article provides a systematic review of the state-of-the-art in PIML, focusing on the methodologies of integrating physics into ML. The PIML techniques can be grouped into three categories, including hybrid models, physical loss-based models, and physics-embedded architectures. Each of these categories is further stratified according to different integration approaches and ML models. The methods and applications of each technique are summarized. In addition, the technical challenges and potential opportunities of PIML are critically analyzed and discussed, providing a roadmap to narrow the research gaps in PIML.
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Keywords: physics-informed machine learning, scientific machine learning, physics-informed neural networks, graph neural networks, Gaussian process,generative adversary network, variational autoencoder, diffusion model, symbolic regression, operator learning, dictionary learning

1 Introduction

Machine learning (ML) has emerged as a widely utilized technique in science and engineering to extract features from data, reveal nonlinear relationships within large datasets, or generate predictive models. When used for the prediction purpose, an ML model is trained based on sampled data, and its structure and parameters are adjusted through iterative processes, without requiring prior knowledge. ML has been extensively applied in design and manufacturing, with various applications such as design optimization, simulation, virtual prototyping, quality control, manufacturing process monitoring, process optimization, process control, robotics, and automation. However, one of the main challenges in design and manufacturing applications is the availability of large training datasets. Complex ML models with a large number of parameters require a significant amount of training data to avoid overfitting. Yet, experimental measurements and simulations in engineering can be very expensive. Another challenge of ML models is their limited extrapolation capability and interpretability in predictions. To improve the accuracy and trustworthiness of ML model predictions, one growing trend is to make "greybox" ML models instead of traditionally "blackbox" ones. When ML models can incorporate prior knowledge of the specific problems, the training efficiency and prediction accuracy can be improved.

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In the past three decades, researchers have developed various strategies to embed physical models and domain knowledge into ML models. These strategies are nowadays referred to as *physics-informed machine learning* (PIML). There are three general approaches to embed physical models and domain knowledge. The first approach is to construct hybrid models that combine ML with physical models sequentially or in parallel for improved prediction [1], or to create physics-informed datasets [2]. The second approach is to use physical models as constraints to guide the training process of ML models [3–5]. The third approach focuses on the development of new ML architectures that leverage problem-specific characteristics and incorporate physics-based guidance to enhance interpretability and ensure physical consistency [6–9]. In these approaches, physical knowledge has been utilized and integrated in ML models through different forms and representations, such as synthetic data, mathematical models, prior experiences, and physical rules. In the literature, these hybrid approaches of integrating domain knowledge in data-driven modeling are also referred to as *scientific machine learning*. Figure 1 systematically summarizes the constitution of existing PIML techniques (figures of prediction and simulation are adapted from Ref. [10]).

The integration of physical knowledge with ML improves the reliability and transparency of the ML models and the accuracy of their predictions. PIML helps reduce the high dependence of ML models on the quality and quantity of input data. However, current PIML techniques are still not capable of accurately modeling and predicting complex problems as traditional physics-based simulations for industry-scale applications. They struggle to solve problems such as modeling the process–structural–property relationship in additive manufacturing, predicting the behavior of materials with phase transitions or nonlinear material responses, and solving multiscale and multiphysics problems [11,12]. Although physics-based models and knowledge have been embedded in PIML models, they do not have the same extrapolation capability as pure physics-based simulations.

To fully understand the strength and weakness of PIML and properly address its bottleneck issues, a comprehensive overview of this research field can help clarify some misunderstanding and reveal the crucial limitations of PIML techniques. Although several articles have already reviewed the recent development of PIML [13–18], they focused on the potential applications of PIML without revealing the key challenges of the PIML methodologies. Good strategies for overcoming these challenges are needed in the research community. Furthermore, most articles devote the same attention to all PIML techniques, without considering the nuances of their impacts on the interpretability of ML models. The goals of this article are to highlight the challenges of PIML by providing a complete historical view, summarizing the methodologies developed in conjunction with different ML approaches, and identifying the common limitations, as well as to offer some prospects of future research opportunities.

The typical physics integration methods and their related advanced techniques in the three categories of PIML methods—namely, hybrid models, physical loss-based models, and physics-embedded architectures—are expounded separately. The advantages and disadvantages of each PIML technique are also examined, accompanied by examples of their applications. By further revealing the existing challenges and potential opportunities of PIML, this article establishes a foundation for researchers to better explore, utilize, and develop PIML techniques. The remainder of this article is organized as follows. Section 2 summarizes the development and applications of hybrid model techniques. Section 3 reviews the methods and physical sources of constructing a physics-based loss function. The advanced techniques developed to improve the performance of physics-loss-based models are also introduced. The details of physics-embedded architectures, including typical model-specified techniques and generic techniques for neural networks, are comprehensively presented in Section 4. The challenges and potential opportunities for PIML are highlighted in Section 5. Section 6 gives the concluding remarks.

2 Hybrid Models

Constructing hybrid data-driven physics-based models is the simplest way to introduce physics into ML. Some hybrid models

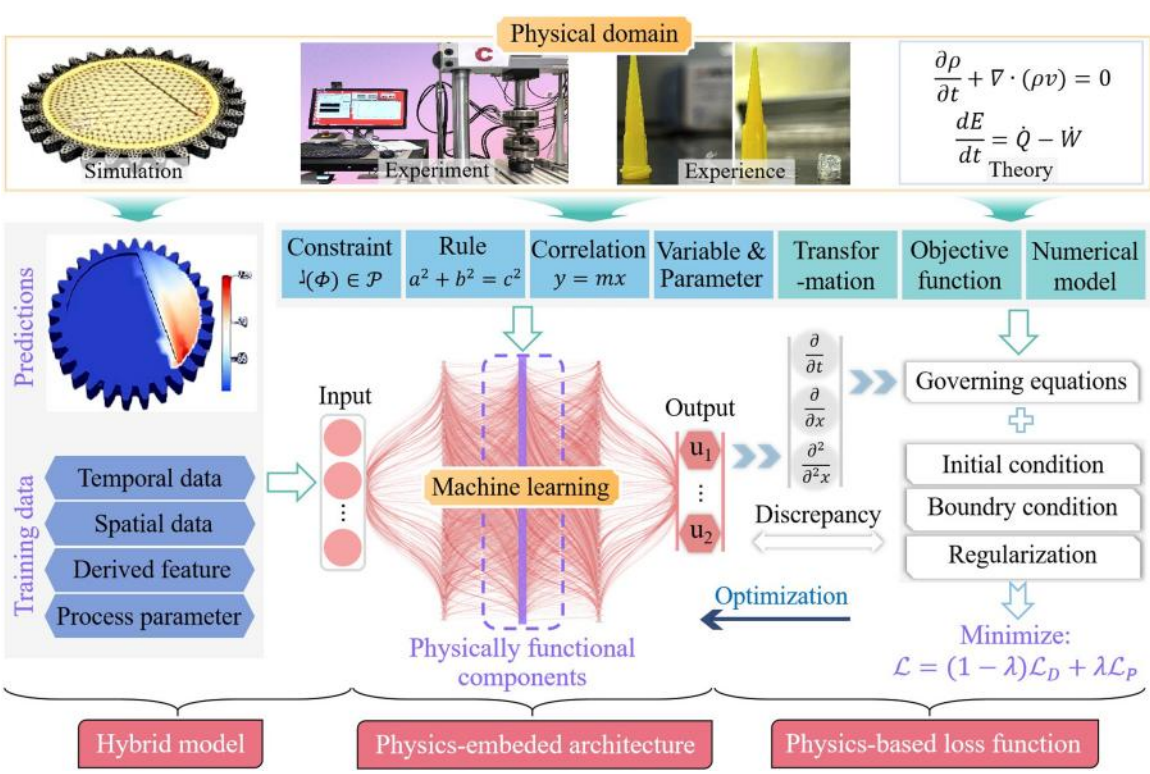

Fig. 1 Overview of PIML techniques

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were already developed in the 1990s to apply ML to solve design and manufacturing problems. This technique involves integrating physical models with ML models in parallel or serial to realize complementary strengths. In other words, the physical model and the ML model operate as independent modules.

2.1 Residual Modeling. The first technique in hybrid models is residual modeling, where ML models are trained to predict errors or residuals of physical models so that the prediction accuracy of physical models can be further improved [1]. That is, when the physical model $\hat{y} = f(x)$ predicts some quantity y, the prediction error $\epsilon(x)$ also depends on input x, where $y = \hat{y} + \epsilon(x)$. The ML serves as the surrogate model of $\epsilon(x)$, as illustrated in Fig. 2.

The ML models adopted in residual modeling are usually some linear regression models. Residual modeling has been widely applied to correct the predictions produced by reduced-order models (ROMs) in dynamical systems and some governing equations. ROMs refer to physical models with reduced complexity compared to the real-world design, and their performance is significantly compromised due to truncation. Guo and Hesthaven [19] adopted residual modeling to approximate parametrized time-dependent problems. They utilized proper orthogonal decomposition (POD) to construct the reduced basis of full-order solutions, and modeled error surrogates using Gaussian process regression (GPR). In model order reduction of thermal fluids, San and Maulik [20] trained a feed-forward neural network (FNN) to account for residual effects of using POD bases to discard Boussinessq equations. The proposed method achieved excellent performance in effectively stabilizing temporal mode evolution. In turbulence modeling, Wang et al. [21] employed random forest to model the errors in Reynolds stresses predicted by Reynolds-averaged Navier–Stokes (RANS) equations. The prediction errors were modeled as functions of mean flow features derived from the RANS results.

2.2 Intermediate Quantity. ML can also be applied to construct intermediate variables or inner components of physical models. That is, the ML model $x = \mathcal{F}(z)$ is constructed to predict the physical quantity x from the observable z. The predicted x is then used as the input of the physical model $f(x)$, as illustrated in Fig. 3.

Intermediate quantity technique contributes to improving the representation and computational accuracy as well as the flexibility of physical models in analyzing complex systems. For example, Parish and Duraisamy [22] adopted GPR to construct the variables representing deficient physics in the turbulence closure model. This method improved the predictive capability of RANS solvers in predicting turbulent channel flow. In geodynamics modeling, Goldstein et al. [23] integrated a predictor derived from genetic programming for oscillatory flow ripples into a morphodynamic model of inner-shelf sorted bedforms. The developed hybrid model successfully achieved parameterizations and captured two pattern modes of sorted bedforms that were previously absent from the model. Similarly, artificial neural networks (ANNs) were used to fit component functions in a high-dimensional model representation to build multidimensional potentials [24]. The neural network procedure not only enables the merger of modal terms to reduce computational costs but also facilitates the determination of high-order modal terms, thereby improving the quality of potential representation. This method achieves a root mean square error of less than $10\,\mathrm{cm}^{-1}$ for fitting and an error of within $1\,\mathrm{cm}^{-1}$ for the calculation of the energy level of the hydrogen peroxide surface.

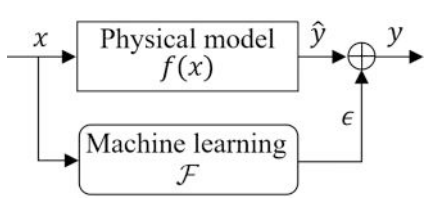

Fig. 2 Residual modeling

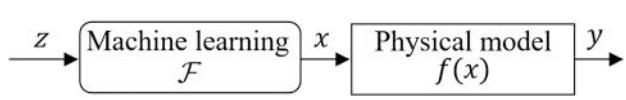

Fig. 3 Intermediate quantity

2.3 Physics-Based Feature Engineering. Physics-based feature engineering is the third technique in hybrid models, which aims to inform the input of ML with physical meaning. It is also the most common technique in hybrid models. Physics-based feature engineering uses the outputs of physical models as the inputs of ML models to predict high-fidelity quantities [2], similar to multifidelity modeling. Specifically, ML model $\mathcal{F}$ can be constructed as $y = \mathcal{F}(x, f(x))$ with both x and $\hat{y}$ as inputs to predict the physical quantity y, as shown in Fig. 4. Physics-based feature engineering helps to promote ML model's exploration of deep physical relationships with limited experimental data. Thus, this technique is an alternative way to address the difficulty of acquiring large amounts of specified data needed for ML through experiments.

Physics-based feature engineering can be achieved through two approaches: physical feature augmentation and physics-based preprocessing. The simplest way to achieve physical feature augmentation is to use simulation models to generate an ML training dataset [25–27]. These simulation models are typically constructed based on governing equations or system models, enabling the generation of synthetic data aligning with the underlying physics. Finite element models (FEMs) are commonly used for generating physical data in applications involving solid structures, where the system's physics is directly incorporated through mathematical equations [28]. For example, in metal additive manufacturing, FEM was used to generate geometrical features of the melt pool, enabling the improved visual geometry group to construct process–property relationships [29]. In structural health monitoring, FEM was adopted to generate structural data of steel beams for training a convolutional neural network (CNN) [30] and to form fault vibration signals of gearboxes for input into an extreme learning machine [31]. Except for FEM-related works, other typical simulation models that have been used to generate physical synthetic data include the phase field model, the fast Fourier transform-based crystal plasticity model [32], the pseudo-two-dimensional model [33], and the diode model [34]. Tian et al. [29] proved that simulation-based feature augmentation can improve the prediction accuracy of ML compared with pure data-driven ML models. However, to ensure reliability of the obtained physical features, the constructed simulation models should be high-fidelity, which imposes stringent requirements on researchers' expertise.

Physical features can also be directly generated through some empirical models and theoretical equations. For instance, Hao et al. [35] employed the Basquin model of the S–N curve, Neuber's rule, and the Smith–Watson–Topper model to generate the reference life of unnotched specimens, the maximum stress at

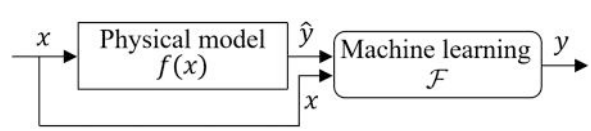

Fig. 4 Physics-based feature engineering

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the notch root, and an energy-type damage parameter for evaluating notch fatigue in aerospace alloys. A prediction accuracy of more than 97% and the improved generalization ability of adopted traditional ML classifiers were achieved. Similarly, Wang et al. [36] applied Paris' law to generate the fatigue crack growth life of materials in additive manufacturing to predict fatigue life, which led to improved fatigue extrapolation and robustness of ANN and support vector regression.

Physics-based preprocessing refers to applying some physical laws or principles to process raw data or extract features that are most pertinent and significant to the problem. One typical approach for the former is to employ some transformations constructed from physical principles or theory equations to map the raw data into physically meaningful features. As described by Xie et al. [37], the wavelet transform was applied to convert infrared thermal histories into frequency-time plots. The transformed features improved the predictive capability of the CNN for location-dependent mechanical properties of additively manufactured parts, even when using limited and noisy experimental data. Other transformation models reported for physics-based preprocessing include logarithmic transformations [38] and the Fourier transform [39]. For the work of physics-based feature extraction, Tian et al. [40] developed a method based on spectral kurtosis and cross-correlation to extract fault features from vibration signals of motor bearings. The cross-correlation model was used to select subsignals containing the fault characteristic frequency, while the spectral kurtosis was used to generate fault features from these subsignals. This strategy enabled the detection of intermittent and hidden faults in motor bearings using K-nearest neighbor distance analysis. The outline of physics-based feature engineering methods can be summarized in Fig. 5.

3 Physics-Based Loss Functions

Unlike the hybrid model techniques in which physics has no effect on ML's training process, physics-based losses are to enhance the training, tuning, and evaluation of ML with physics. Physical knowledge, such as governing equations and mathematical relationships, is incorporated into the loss function of ML models as the physical loss. Through backpropagation updates, the physical loss can guide ML models to capture the patterns that align with the defined physical laws. The idea of formulating physics-based loss to train ML models can be traced back more than two decades ago. In 1998, Lagaris et al. [3] proposed to use an ANN to solve differential equations $y = f(x, \psi(x), \nabla\psi(x), \nabla^2\psi(x))$ by formulating the loss function of neural networks as

$$\mathcal{L}_P = \|y - f(x, \psi(x), \nabla\psi(x), \nabla^2\psi(x))\|^2 \tag{1}$$

where y is the output of neural networks, $f(\cdot)$ is a general operator of differential equations, and $\psi(x)$ is the solution to be computed. Lagaris et al. [3] demonstrated this approach with several ordinary differential equations (ODEs) and partial differential equations (PDEs). In 2007, Cursi and Koscianski [4] extended the approach by combining data loss $\mathcal{L}_D$ and physical loss $\mathcal{L}_P$ with the Lagrange multiplier λ as

$$\mathcal{L} = (1 - \lambda)\mathcal{L}_D + \lambda\mathcal{L}_P \tag{2}$$

That is, the physical loss is the regularization and serves as a soft constraint to guide the training of neural networks.

This concept of physics-based loss function has recently been revived and has gained much more attention. One of the most popular methods is a physics-informed neural network (PINN) [5], also known as a physics-constrained neural network (PCNN). When the physics-based loss function approaches zero, it indicates that the predictions satisfy both the physical laws and the labeled data simultaneously [41]. Physical knowledge is thereby integrated into the model [42]. These physical laws range from simple formulas to complex PDEs [13,43]. The following sections will detail the different methods of constructing physical loss and the related techniques for improving the performance of physics-loss-based PIML models.

3.1 Loss Function by Automatic Differentiation. Neural networks are designed to use spatiotemporal coordinates as inputs and produce variables of interest as outputs [44]. This process generates a computational graph that maps the inputs to the outputs. During backpropagation, the gradients for each node within the graph must be computed [45]. Automatic differentiation (AD) technique applies the chain rule to evaluate derivatives of arbitrary orders, thereby obtaining the temporal and spatial derivatives of the variables of interest [46], as shown in Fig. 6. These derivatives are subsequently used to construct the physical loss functions in accordance with the target PDEs.

Since derivatives at any point within a domain can be easily obtained by AD, this type of loss function can be readily adapted to various problems [47,48], whether they involve regular or irregular geometries [49–51]. In heat analysis, such loss functions can address both forward and inverse problems to obtain the temperature distributions and predict thermal properties [44,52]. In fluid dynamics, they help predict flow fields or reconstruct missing information with available measurements [53]. Furthermore, in multiphysics modeling, thermofluid dynamics in the melt pool of metal additive manufacturing can be monitored using AD-based PINNs [54]. Conservation laws of momentum, mass, and energy during the additive manufacturing process were used to model the three-dimensional (3D) melt pool [55]. The prediction discrepancy (with respect to the experimental measurements) of PINN was decreased by 14.2% compared to numerical simulations, even with a moderate amount of labeled datasets. In addition, they have been applied to analyze interactions between blood flow and vessel wall mechanics [49,56]. Arzani et al. [57] developed a PINN-based framework to accurately quantify near-wall blood flow and wall shear stress in diseased arterial flows using sparse measurements and partial boundary conditions. Figure 7 illustrates the framework of AD-based PINNs with alternative neural network models.

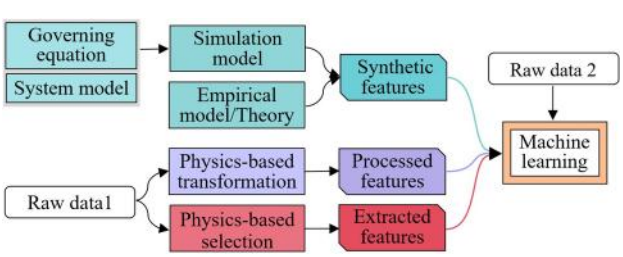

Fig. 5 Outline of physics-based feature engineering methods

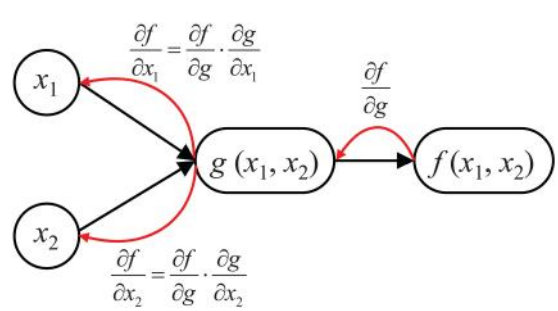

Fig. 6 Schematic diagram of AD including forward accumulations and gradient computations

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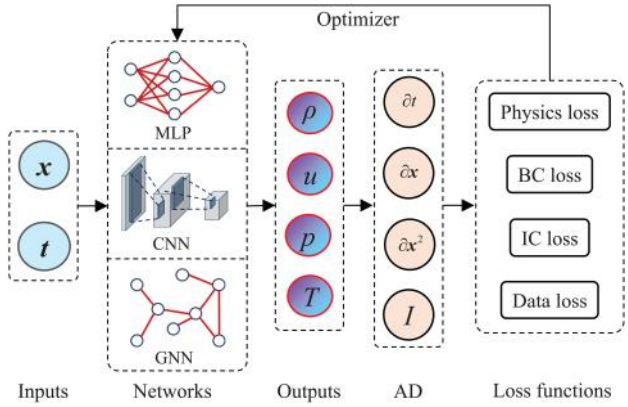

Fig. 7 Framework of AD-based PINNs with various neural network architectures

3.2 Loss Function by Numerical Discretization. In addition to AD, physical loss functions can also be constructed based on numerical discretization of PDEs [58]. Common numerical methods include the finite difference method (FDM) [59], the finite volume method (FVM) [60], and the FEM [61]. This approach requires neural networks to generate predictions on a predefined grid, ensuring compatibility with the discrete domain representation [62]. By leveraging these numerical techniques, the outputs from neural networks are directly used to construct the various derivatives in the target PDEs, which avoids the lengthy chain of AD. The framework of PINNs incorporated with numerical methods is shown in Fig. 8.

Recently, a physics-informed fully convolutional network (FCN) was developed by employing the FDM to construct a physical loss function for a parametric heat conduction equation [63]. This model can simultaneously predict the temperature field and estimate unknown properties. The results for forward temperature prediction and inverse estimation of the unknown thermal diffusivity are shown in Fig. 9. The maximum error of the predicted temperature is 1.47×10^{-2}, and the accuracy of the estimated thermal diffusivity is 99.3%. Chiu et al. [64] compared their proposed hybrid network with numerical differentiation to AD-based PINNs and found that its accuracy exceeded that of the traditional one in various problems. Ranade et al. [65] proposed a ML-based solver for the Navier–Stokes equation using FVM. The experiments showed that the computation accuracy, stability, and convergence speed of the proposed solver are significantly improved compared to the ANSYS Fluent solver. Additionally, Zhu et al. [66] developed a finite volume physics-informed U-net (FV-PIUnet) by integrating FVM into the U-net and applied this approach to resolve Burgers' equation. The results in Fig. 10 illustrate that FV-PIUnet achieves higher accuracy compared to an AD-based PINN, with a maximum error of 0.37 for FV-PIUnet and 1.76 for the AD-based PINN. Yadav and Ganesan [67] combined FEM and a neural network to solve a two-dimensional (2D) convective-diffusion equation. Their approach suppressed the undershoots and overshoots that were encountered in the existing neural network-based PDE solvers. These studies demonstrated that the significant drawbacks of AD in constructing physical loss can be effectively alleviated.

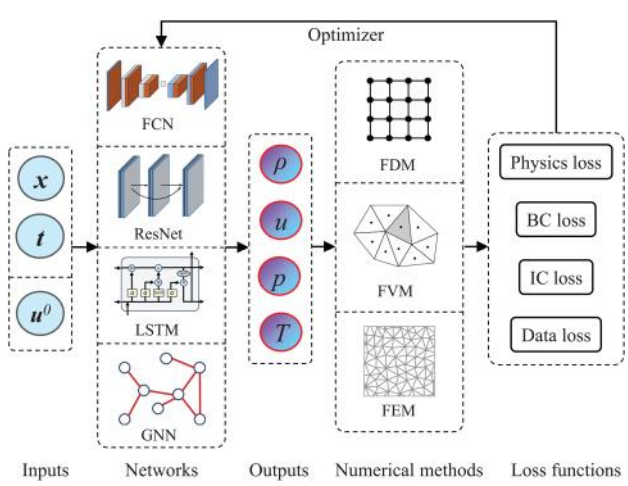

Fig. 8 Framework of PINNs with various neural network architectures combined by numerical methods

Combining PINNs with numerical loss functions can be used as surrogate models for traditional numerical approaches to solve linear and nonlinear PDEs [65,68]. By training with initial conditions or early stage solutions, the surrogate model can effectively extrapolate the solution over the untrained period [69,70]. This approach proves particularly advantageous in addressing high-dimensional stochastic PDEs, such as solving fluid flow through random heterogeneous media. Zhu et al. [71] proposed a physics-constrained surrogate model for the Darcy flow problem. Their model does not require labeled data and achieves results comparable to numerical simulations. In such cases, the surrogate model is capable of efficiently handling and solving the equations.

3.3 Additional Regularization Term. Although integrating loss terms based on governing PDEs into the total loss function facilitates the neural network to learn physical laws, there are still challenges in prediction accuracy and convergence. Consequently, some studies have introduced additional loss terms, also known as regularization terms, to address these issues [72,73].

The regularization terms include two categories. The first one involves regularizing the individual weights when the loss terms are combined by weighting. That is, an additional regularization term is added in the loss function to constrain the magnitudes of the individual weights to prevent any negative impact on individual losses [74]. The second category is to construct regularization terms using additional physical information or constraints. For example, when solving inviscid Burgers and Euler equations in compressible flow problems, introducing artificial viscosity in the original PDEs facilitates faster convergence [75], which can be expressed as

$$L_{\mathrm{av}} = \frac{1}{n}\sum_{i=1}^{n}\left(\partial_t \hat{u}_i + \nabla f(\hat{u}_i) - \hat{\mu}_i \nabla^2 \hat{u}_i\right)^2 \tag{3}$$

where $\hat{u}$ is the output of the neural networks, f is the flux function, and $\hat{\mu}$ represents the artificial viscosity coefficient. In solving heat transfer problems, the incorporation of entropy inequalities and enthalpy conservation relations can further enhance the neural network's ability to learn physical laws [76]. The loss function for entropy inequality is given by

$$\mathcal{L}_{\mathrm{ent}} = \sum_{i=1}^{N}\left(\max\left(0, \int \hat{\mathbf{q}}_i \cdot d\mathbf{A}\right)\right)^2 \tag{4}$$

where $\hat{\mathbf{q}}$ denotes the entropy flux, and $\mathbf{A}$ represents the differential area of the grid cell. Moreover, when constructing physical loss functions using numerical methods, additional numerical scheme constraints can be introduced as the regularization term. For instance, Patel et al. [76] utilized total variation diminishing (TVD) loss $\mathcal{L}_{\mathrm{TVD}}$ to ensure that new oscillations or spurious variations are not introduced into the predicted solutions, as

$$\mathcal{L}_{\mathrm{TVD}} = \sum_{n} \max(0, TV_h(\boldsymbol{u}_{n+1}) - TV_h(\boldsymbol{u}_n))^2 \tag{5}$$

where the discretized form of TV_h is given by

$$TV_h(\boldsymbol{u}_n) = \sum_{i} \left|\boldsymbol{u}_{i+1,n} - \boldsymbol{u}_{i,n}\right| \tag{6}$$

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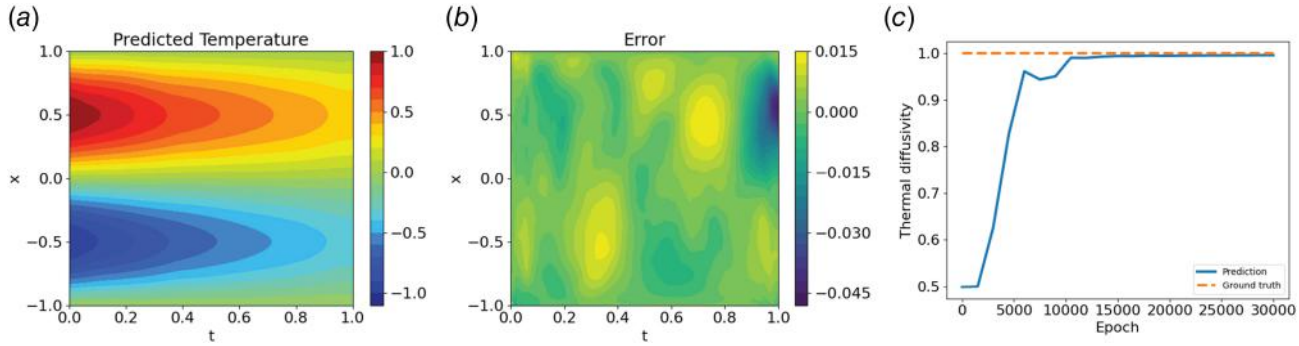

Fig. 9 (*a*) Predicted temperature fields, (*b*) error distributions, and (*c*) inverse solution of the thermal diffusivity [63]

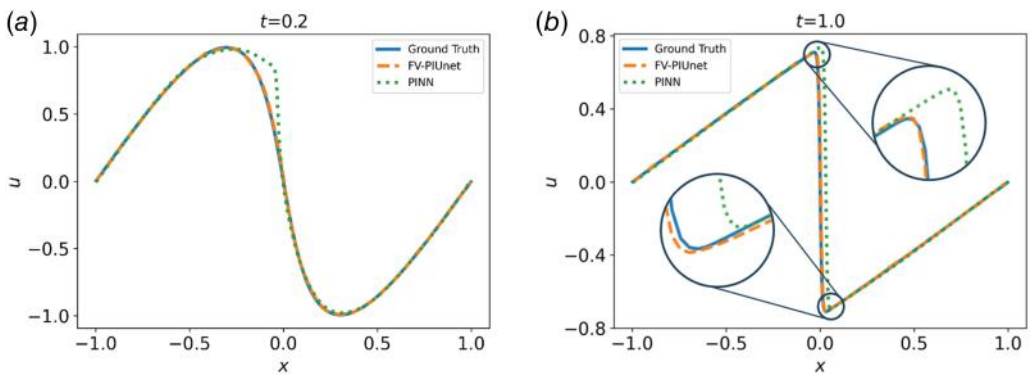

Fig. 10 The comparison of velocity profiles at (*a*) nonshock region ($t = 0.2$) and (*b*) shock region ($t = 1.0$) [66]

Other examples of regularization have also been developed. For instance, the regularization based on homotopy continuation allows for tracing the observations and identifying multiple solutions of PDEs [77]. Grouping losses locally in different regions can alleviate the challenge of magnitude difference [78]. A regularization that penalizes the unstable solutions based on the spectrum of the Jacobian matrix can avoid unstable fixed points in dynamical problems [79].

3.4 Weighting Strategy. Training PINNs is a multitask learning process because the total loss function of a PINN often involves the components of sampling data loss, initial and boundary condition (I/BC) loss, and PDEs loss. All of the sublosses are expected to converge simultaneously. However, significant differences between the magnitudes of these losses hinder the optimizer to converge all loss terms [80]. Therefore, it is necessary to combine them into a total loss function using a weighted approach. To rationalize the contributions of all the loss terms, various weighting strategies have been developed to appropriately adjust the weights. Liu and Wang [11] first proposed an adaptive weighting scheme to adaptively adjust the weights of PINN. For each epoch of training, the weights are adjusted based on the respective magnitudes of physical losses. The effectiveness of this approach was demonstrated in solving multiphysics problems such as heat transfer and the Allen-Cahn equation in dendritic growth. Another self-adaptive loss balanced method is based on the maximum likelihood estimation which was proposed by Xiang et al. [81]. With this loss-balance strategy, the robustness of PINN is improved against variations in initial weights, the number of observations, hidden layers, and neurons per layer. However, in most PINN-based research, the weights of losses are only manually tuned. For example, Wight and Zhao [82] applied a large fixed weight for the loss of initial conditions to improve overall accuracy when solving the Allen-Cahn and Cahn-Hilliard equations. The manually tuned weights rely entirely on experience and can vary from case to case.

In the adaptive weighting scheme, the total loss function for model training is defined as

$$\mathcal{L}(\mathbf{w}) = \lambda_D \mathcal{L}_D(\mathbf{w}) + \lambda_P \mathcal{L}_P(\mathbf{w}) + \lambda_I \mathcal{L}_I(\mathbf{w}) + \lambda_S \mathcal{L}_S(\mathbf{w}) \tag{7}$$

where $\mathcal{L}_I$ and $\mathcal{L}_S$ represent the losses arising from violations of the initial and boundary conditions, respectively. The weights of different losses λ_D, λ_P, λ_I, and λ_S satisfy the constraint $\lambda_D + \lambda_P + \lambda_I + \lambda_S = 1$.

The adaptive scheme involves assigning weights to the different losses, such as

$$\lambda_i = \frac{\mathcal{L}_i}{\mathcal{L}_D + \mathcal{L}_P + \mathcal{L}_I + \mathcal{L}_S}, \quad i \in \{D, P, I, \mathcal{S}\} \tag{8}$$

for each iteration of the training process. In other words, the weights are assigned in proportion to the respective individual losses.

To further improve the convergence, Liu and Wang [83] further proposed a new minimax architecture for PINN with accelerated weight adjustment. The training of PINN becomes solving minimax problems. The optimization objective is given by

$$\begin{aligned}\min_{\mathbf{w}} \max_{\boldsymbol{\hbar}} \mathcal{L}(\mathbf{w}, \boldsymbol{\hbar}) = \lambda_D(\boldsymbol{\hbar})\mathcal{L}_D(\mathbf{w}) + \lambda_P(\boldsymbol{\hbar})\mathcal{L}_P(\mathbf{w}) \\ + \lambda_I(\boldsymbol{\hbar})\mathcal{L}_I(\mathbf{w}) + \lambda_S(\boldsymbol{\hbar})\mathcal{L}_S(\mathbf{w})\end{aligned} \tag{9}$$

where the weights of different losses λ_D, λ_P, λ_I, and λ_S are now functions of parameters $\boldsymbol{\hbar} = (\hbar_D, \hbar_P, \hbar_I, \hbar_S)$. Training is to minimize the possible loss for a worst-case (maximum loss) scenario. That is, the total loss $\mathcal{L}(\mathbf{w}, \boldsymbol{\hbar})$ is maximized in the parameter subspace of $\boldsymbol{\hbar}$ while minimized in the parameter subspace of $\mathbf{w}$. During the training, the weights of different losses λ's are adjusted to maximize the total loss $\mathcal{L}(\mathbf{w}, \boldsymbol{\hbar})$ in $\boldsymbol{\hbar}$ subspace, whereas the

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weights of the neural network $\mathbf{w}$'s are tuned to minimize the total loss $\mathcal{L}(\mathbf{w}, \boldsymbol{\hbar})$. The weights of different losses can be defined as the softmax function

$$\lambda_i(\hbar) = \frac{\exp(\hbar_i)}{\exp(\hbar_D) + \exp(\hbar_P) + \exp(\hbar_I) + \exp(\hbar_S)}, \quad i \in \{D, P, I, S\} \tag{10}$$

The applications in solving a heat transfer problem demonstrated that the computational efficiency of the minimax architecture is higher than that of the gradient descent method in traditional PINNs.

3.5 Gradient Improvement Method. When training PINNs/PCNNs to solve PDEs, many factors related to gradient computation may lead to poor performance. Common factors include gradient imbalance [84], gradient pathology [85], multitask conflicts [86], and uneven gradient distribution caused by multiscale problems [13]. The gradient improvement method helps alleviate these problems by improving the gradient calculation process, such as optimizing the direction or path of the gradient update [87].

Researchers have developed many gradient improvement methods. Gradient balancing is one of the most representative approaches, which aims at addressing gradient imbalance problems among multiple loss terms. It can be achieved by dynamically adjusting the weights based on the gradient norm [84]. The principle of this method is formulated as

$$\lambda_i^{(t+1)} = \frac{\overline{|\nabla_\theta \mathcal{L}|}}{\max(|\nabla_\theta \mathcal{L}_i^{(t)}|, \varrho)} \tag{11}$$

where $\lambda_i^{(t+1)}$ denotes the weight of the ith loss term at the $t+1$ iteration. $\overline{|\nabla_\theta \mathcal{L}|}$ refers to the mean absolute value of the gradients of all the loss terms. θ is the network parameters. ϱ is a small positive value to avoid a zero denominator. Gradient balancing can also be achieved through gradient normalization, which involves rescaling the gradients of each loss term to ensure consistent magnitudes across tasks [85]. Another effective gradient balancing method is task prioritization [13], which balances the gradient by adjusting the optimization steps to compensate for the gradient difference. Specifically, it uses an adaptive sampling strategy to increase the sample points in regions where the gradients are smaller, such as when the PDE residual is higher.

The second method to improve gradient propagation is the optimization of the gradient path. It involves constraining the gradient path or using advanced optimizers to improve optimization trajectories, thereby avoiding local optima and slow convergence [84]. Gradient clipping and gradient projection are common methods for constraining the gradient path. Specifically, gradient clipping stabilizes the optimization process by limiting the maximum norm of gradients to avoid the gradient explosion issue. Gradient projection projects the gradients into directions that satisfy the targeted physical or mathematical constraints. This ensures that the parameter update process satisfies the defined constraints. Thus, the invalid gradient update can be avoided [86]. In the aspect of employing advanced optimizers, Adam with decoupled weight decay (AdamW) [88] and limited-memory Broyden Fletcher Goldfarb Shanno (L-BFGS) [89] are the commonly used methods. The AdamW optimizer reduces the oscillation of the gradient path by introducing weight decay regularization parameters. The L-BFGS uses second-order information (Hessian approximation) to calculate a better descent direction [13].

Gradient regularization is the third technique that can avoid gradient pathology and improve gradient stability. The related techniques include gradient norm regularization and high-order gradient regularization. The former functions by penalizing large gradients, while the latter aims to smooth the high-order derivatives of the network output [13,84]. Complementary to regularization, gradient decomposition techniques address the issue of conflicting gradients in multitask optimization. For instance, projecting conflicting gradients (PCGrad) [90] resolves gradient conflicts by projecting the gradient of one task onto the normal plane of another. Thus, it can eliminate incompatible components and enable more consistent optimization [84]. In terms of gradient computation, the adjoint method [91] improves efficiency by decomposing the total gradient into a forward pass and an adjoint (backward) pass. This method avoids direct computation of all intermediate derivatives, allowing scalable and memory-efficient training, which is especially useful when computing high-order derivatives in PINNs.

3.6 Sampling Strategies for Residual Point. Sampling residual points in the computational domain is critical for constructing AD-based physical loss functions, as it directly affects convergence behavior, solution accuracy, and computational cost. Conventional strategies, such as random sampling (e.g., Latin hypercube sampling) and uniform sampling, may lead to large local errors in strongly nonlinear regions or where gradients vary rapidly. Consequently, various adaptive sampling strategies have been proposed. An intuitive approach is to repeatedly sample according to the error distribution. Specifically, residual points are first randomly selected and used to train PINNs, and their locations are then adjusted according to the predictions of the model. This process can be repeated multiple times to obtain a more refined set of residual points. For instance, Cen and Zou [92] conducted 10 rounds of resampling, resulting in a substantially lower L^2 relative error compared to models trained without this technique. However, the training time increased nearly fivefold.

Another approach for distributing residual points is to leverage the patterns of boundary or initial conditions, which can significantly improve the prediction accuracy. Since boundary and initial conditions have a substantial impact on subsequent solutions, increasing sampling density in these regions is highly beneficial. For instance, Zhao et al. [93] sampled residual points based on the initial condition when solving one-dimensional (1D) and 2D Euler equations. Their results demonstrated accurate predictions of both the initial and final states. The gradients at the interface are also effectively captured. In addition, the nature of the governing PDEs can guide the placement of residual points. In hyperbolic PDEs, shock formation is closely related to the intersection of characteristic lines, making it helpful to add more sampling points around them to ensure accurate shock prediction. Mao et al. [50] employed a uniform distribution of residual points and increased sampling density along characteristic lines, yielding a sharper shock wave. Similarly, Jagtap et al. [94] precomputed the positions of forward and rearward Mach lines, and then added more sampling points near these lines in an otherwise randomly distributed set, leading to improved shock predictions. Compared with traditional methods, both of these strategies necessitate additional prior knowledge, which will pose challenges in high-dimensional or complex problems.

3.7 Advanced Architecture-Based PINN. Traditional PINNs are primarily built upon multilayer perceptrons (MLPs), leading to low computational efficiency. To address this issue, advanced architecture-based PINN techniques have been developed, which aim to replace MLPs with more advanced architectures. For example, fully connected layers can be changed into convolutional layers to reduce connection densities. Some studies have applied FCNs and U-Net architectures to solve problems related to fluid flow and heat transfer, yielding shorter training times and higher accuracy compared to standard PINNs [58,64,95]. A variational autoencoder (VAE) has been integrated to represent complex geometries of the domain [96]. Moreover, in certain surrogate models for PDE solvers, recurrent neural network (RNN) and long-short term memory (LSTM) structures have also been explored, underscoring the potential for even broader design choices in the development of enhanced PINN

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frameworks [97,98]. Tong et al. [70] presented a novel network inspired by the classical Roe solver to predict the solutions of hyperbolic PDEs. Comparisons with the original solver revealed that the neural network-based approach exhibited slight superiority in shockwave shape and position. Ren et al. [68] developed a physics-informed convolutional LSTM to solve the nonlinear convective PDEs. Their method outperformed the original one in terms of solution accuracy, extrapolability, and generalizability. Peng et al. [99] developed a graph neural network (GNN) that served as a prediction module. They extended the geometric adaptability to irregular domains by employing unstructured meshes. The predicted velocity fields of fluid flow around a cylinder using the proposed model closely aligned with the ground truth, even with very limited training data. Table 1 summarizes the typical literature on advanced architecture-based PINN, with an emphasis on their application cases, merits, and limitations.

3.8 Multifidelity PINN. Although PINNs require less training data compared to pure data-driven neural networks, their accurate predictions still rely on the input of sufficient high-fidelity data [74]. However, high-fidelity data usually imply increased experimental complexity and acquisition challenges, which may not be available in all scenarios. The multifidelity PINN technique was specially developed to reduce the dependence of traditional PINNs on large amounts of high-fidelity data while ensuring high computational accuracy.

The principle of multifidelity PINN is to simultaneously sample data with various fidelity levels during training, which is similar to the concept of multifidelity surrogate modeling [108–110]. One way to achieve this is to integrate multiple neural networks of different fidelity, which is also referred to as a multifidelity physics-constrained neural network with a minimax architecture (MF-PCNN-MM) [74]. The MF-PCNN-MM is able to improve prediction accuracy and training stability by physical constraints and minimax optimization, while also lowering the costs associated with high-fidelity data by utilizing low-fidelity data. Figure 11 shows the architecture of MF-PCNN-MM, which consists of two subnetworks.

The first subnetwork makes predictions based on low-fidelity data, and its output $U_L(x)$ is used as the partial input of the second subnetwork. The second subnetwork $U_H(x, U_L)$ approximates the correlation function between the low-fidelity and high-fidelity data, which can be expressed as [108]

$$u_H(x, u_L) = \rho_s(x)u_L + \delta(x) \tag{12}$$

where x is the input vector. u_L and u_H are the predictions of the low-fidelity model and the high-fidelity model, respectively. $\rho_s(x)$ represents the multiplicative correlation surrogate, and $\delta(x)$ is the additive correlation surrogate. That is, the high-fidelity network $U_H(x, U_L)$ serves as a surrogate model for the left side of Eq. (12). The training process of the multifidelity physics-constrained neural network (MF-PCNN) is formulated as a minimax optimization problem using the method in Eq. (9).

During the training of MF-PCNN-MM, a bilevel optimization is performed: the loss weights $\lambda_i(\hbar)$ are optimized to maximize the total loss $\mathcal{L}(w, \hbar)$, while the neural network parameters w are updated to minimize it. This optimization process is equivalent to finding a saddle point in a high-dimensional energy landscape E, where the Dual-Dimer method [83] is used to locate such high-order saddle points.

The proposed scheme is demonstrated with three examples: heat equation, Allen-Cahn equation, and dendritic growth during solidification, which involves coupled heat transfer and phase transition. Across all cases, the proposed scheme consistently achieves higher predictive accuracy compared to the multifidelity neural network with minimax architecture (MF-NN-MM). Moreover, it is able to capture more intricate interfacial morphologies and thermal field evolution of the physical systems.

3.9 Physics-Informed Bayesian Neural Networks. Similar to neural networks, PINNs face the accuracy challenge. Based on the sources of uncertainty, the errors can be categorized as either model-form uncertainty or parameter uncertainty. The model-form uncertainty in neural networks is associated with the choices of network architecture, neuron connections, and activation functions. The choices based on analysts' preferences and

Table 1 Literature summary of advanced architecture-based PINN

References	ML model	Physics loss method	Applications	Merits	Limitations
[5,41,42]	MLP	AD	Forward and inverse problems in PDEs	Concise and intuitive, easy to learn and apply. A meshless approach for multifield physical problems. A unified framework for solving forward and inverse problems.	High computational resource and time-consuming. Not accurate enough in gradient learning, especially for nonlinear PDEs.
[54,100]	MLP	AD	Temperature field and the melt pool fluid dynamics in additive manufacturing		
[58,63,71]	CNN	FDM	High-dimensional surrogate model of stochastic PDEs	High efficiency in gradient computation. Capable of applying hard constraints on boundary conditions to enhance accuracy.	Based on structured mesh, hard to handle complex geometries. High demand for data on boundary conditions. Low flexibility.
[68,101]	LSTM	FDM	Surrogate model		
[70]	ResNet + CNN	FVM	Hyperbolic PDEs	Supporting both structured and unstructured meshes, and compatible with complex geometries. Ensuring flow conservation and providing higher accuracy in convective problems.	Additional computational cost for creating unstructured meshes. Dependency on mesh quality.
[102,103]	GNN	FVM	Incompressible flow and heat transfer		
[62,104,105]	FCN	FEM	Mechanical response and material phases	Effectively handling irregular geometries. A unified approach to multiphysics problems.	Challenges in building loss functions. High training complexity and difficulty in convergence.
[106,107]	GCN	FEM	Forward and inverse problems in thermal analysis, mechanics, and fluid dynamics		

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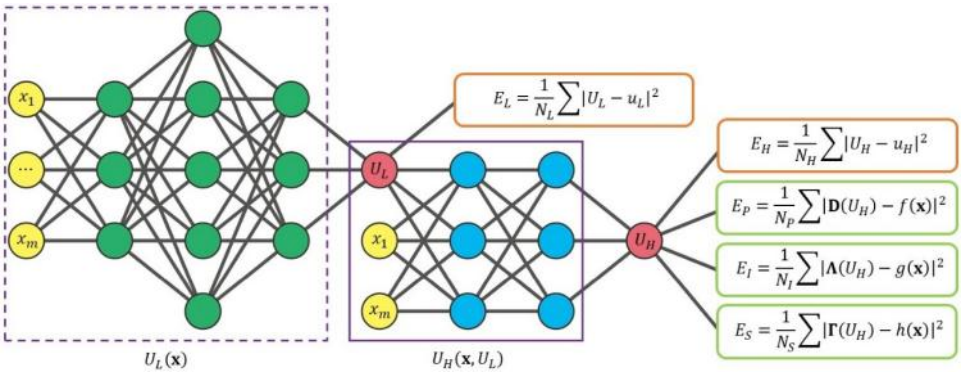

Fig. 11 Framework of MF-PCNN-MM architecture [74]. Reprinted with permission from Liu et al. © 2023 ASME.

own knowledge naturally introduce bias and potential errors into the model predictions. The parameter uncertainty is associated with the training of neural networks, where errors could be caused by the quality and quantity of training data, as well as training algorithms and training time. Training neural networks with a large number of parameters is performing optimization in the high-dimensional parameter space, and it is very difficult to find the global optimum. Suboptimal parameters will lead to errors in the model predictions. For PINNs, model-form and parameter uncertainty are also associated with the physical models. Simplification, approximation, and truncation of mathematical forms involved in physical models lead to model-form uncertainty, whereas the calibration process is the major source of parameter uncertainty. Therefore, the physical models themselves are additional sources of errors.

Several approaches have been developed to quantify the uncertainty in neural networks, including Bayesian neural networks (network parameters follow statistical distributions), Monte Carlo dropout (neuron connections are randomly disconnected), a hybrid neural network Gaussian process model (the output layer of neural networks is replaced with Gaussian process models), and a neural network ensemble (an ensemble of networks that predict variations of quantities) [111]. To quantify the uncertainty in PINNs, physics-constrained Bayesian neural networks (PCBNNs) [112,113] have been proposed. The weights in PCBNNs are probabilistically distributed, instead of deterministic values as in classical PINNs. The training of PCBNN is based on Bayes' rule. The goal is to minimize the difference between the distribution observed in training data and the distribution predicted by PCBNN. In the iterative training process, the prior distribution of neural network weights is updated to the posterior given the likelihood of observing the data. The most generic training algorithm is the Markov chain Monte Carlo (MCMC), where the posterior distribution is obtained by sampling. However, training based on MCMC is computationally very expensive. A more efficient alternative is variational inference. In variational inference, the weights are regarded as the latent variables.

Instead of training and updating the weights of PCBNN, recently a strategy of Monte Carlo dropout was developed. In Monte Carlo dropout, weights in traditional neural networks are randomly chosen and set to zero. As a result, the predictions from the ensemble of neural networks are stochastic. The dropout strategy was originally designed for the sensitivity analysis of neuron connectivities. For uncertainty quantification, Monte Carlo dropout provides an efficient way to assess the effect of model-form uncertainty on neural networks.

4 Physics-Embedded Architectures

Despite the physics-based loss function and hybrid model techniques that enable the search space, input, and output data to be physically meaningful, the ML architecture remains a black-box. To force the black-box to be interpretable and further improve ML performance, researchers have devoted significant efforts to constructing physics-embedded architectures. Due to significant differences in theoretical foundations and architectures among different ML models, the methods for embedding physical knowledge vary. In some early work of PIML, specialized ML architecture designs were developed so that physics information is embedded in interpretable ML structures. For instance, Ramuhalli et al. [6] designed a finite element neural network, where the stiffness matrix elements are directly mapped to the activation functions of neurons so that the influence of material properties can be predicted. Behler and Parrinello [7] constructed neural networks to predict the atomic potential energy, where the activation functions are the radial basis functions that preserve the angular and distance symmetries in atomic systems.

In this section, a detailed overview of the methodologies and applications of several specific physics-embedded ML architectures is provided. Some ML techniques other than neural networks, such as dictionary learning, are also summarized.

4.1 PDE- and Hard-Constrained Neural Network. Unlike soft-constrained approaches that incorporate PDE residuals into the loss function, hard-constrained neural networks directly enforce specific PDE conditions, such as boundary conditions, initial conditions, or conservation laws, through deliberate network architecture design. This strategy minimizes dependence on residual-based penalties, thereby reducing computational costs and enhancing efficiency in solving PDEs. The very early work of PIML by Lagaris et al. [3] explicitly considered the initial and boundary conditions in an analytical form instead of including them in neural networks.

More recently, several approaches were developed to impose hard constraints on the neural network framework, which focus on improving the consistency of predictions with respect to the known conditions. The initial and boundary conditions enforced neural network [114–116] is a specifically designed hard-constrained framework that embeds boundary or initial conditions directly into the neural network architecture through mathematical construction. For example, Straub et al. [117] presented a method to enforce Neumann boundary conditions in PINNs using Fourier feature embeddings. They employed a cosine-only Fourier feature embedding, which ensures that the spatial derivatives vanish at the boundary, thereby hard-enforcing the Neumann condition. Deguchi and Asai [118] showed that boundary conditions can be enforced exactly by employing distance functions, in particular normalized R-functions, which convert the PINN output into a boundary-satisfying trial solution. By doing so, it ensures that the network output strictly satisfies the prescribed constraints along the initial or boundary nodes. A typical embedding form can be

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expressed as

$$y(x, t) = g(x, t) + h(x, t) \cdot N_\theta(x, t) \tag{13}$$

where $g(x, t)$ is a predefined function that satisfies the initial or boundary conditions. $N_\theta(x, t)$ is the output of the neural network. $h(x, t)$ is an auxiliary function, usually designed as zero along the boundary nodes to ensure that the contribution of $N_\theta(x, t)$ is zero along the boundary, or $h(x, t) = t$ when serving as embedding initial conditions.

Conservation law enforcement is a hard-constrained neural network technique that incorporates physical conservation principles, such as mass, momentum, or energy conservation, directly into the network architecture [42]. Typical implementations include divergence-free constraints for incompressible flows ($\nabla \cdot \mathbf{u} = 0$) [42] and the integral conservation of quantities [119]. Négiar et al. [120] introduced an additional layer in the neural network, which computes a linear combination of outputs at different spatial locations. The training is to drive the PDE constraint to be satisfied collectively at all locations, so that the errors at those locations can be reduced compared to the individually constrained PINN model. Min and Azizan [121] enforced hard constraints by appending a differentiable projection layer after the network output that maps predictions onto an input-dependent feasible set (e.g., affine/convex). Thus, it enabled unconstrained end-to-end training while guaranteeing constraint satisfaction. Furthermore, Iftakher et al. [122] proposed a Karush-Kuhn-Tucker (KKT)-HardNet, which introduces a differentiable projection layer. By solving the KKT system of an orthogonal distance-minimization problem, it projects network outputs onto the feasible set and enforces nonlinear equality and inequality constraints. These methods eliminate the need to include explicit residual terms in the loss function, as the network structure inherently satisfies the governing physical laws. Variational formulations with hard constraints reformulate PDEs as functional optimization problems or their integral equivalents, where the loss is defined based on the weak form of the governing equations. The hp-variational physics-informed neural network framework [123] exemplifies this approach by incorporating the PDE residual into the loss through variational integration, thereby enforcing the underlying physics as a hard constraint. A representative loss function in this formulation is given by

$$\mathcal{L} = \int |\mathcal{R}(y_\theta)v|^2 \, d\Omega \tag{14}$$

where y_θ is the output of the neural network, v is the testing function, Ω is the solution domain, and $\mathcal{R}(y_\theta)$ is the residual.

Neural operators also demonstrate strong potential for incorporating hard constraints within their frameworks, aiming to learn solution operators of PDEs that inherently satisfy the governing equations. Hard constraints can be applied to neural operators through mechanisms similar to initial and boundary condition embedding, as well as conservation law enforcement. Representative models such as deep operator networks (DeepONets) [124] and the Fourier neural operator (FNO) [125] have shown the capability to embed such constraints. However, most existing hard constraint techniques have primarily focused on standard neural networks, leaving neural operators a promising direction for future development.

4.2 Physics-Informed Gaussian Process Regression. GPR, also known as Kriging, is a nonparametric statistical model that supports active learning, where its model update is based on the Bayesian approach. The core idea of GPR is to model the unknown function $f(x)$ as a Gaussian process (GP), described by $f(x) \sim \mathcal{GP}(m(x), k(x, x'))$, where $m(x)$ is the mean function of input x, and $k(x, x')$ is the covariance function or kernel function that characterizes the similarity between two inputs x and x'. In contrast to the common supervised-learning workflow that utilizes a fixed training dataset, GPR enables sequential active learning. That is, the current mean and variance functions are used to choose new query locations via an acquisition criterion (e.g., uncertainty level or information gain). The corresponding new observations are collected, and the model is updated with Bayes' rule. The assumption of a Gaussian distribution makes each update computationally efficient because of Gaussian–Gaussian conjugacy, while the kernel's hyperparameters and the noise level are estimated by maximizing the log marginal likelihood or via a fully Bayesian treatment [126].

The original GPR is a generic mathematical model. For specific applications, the GPR surrogate model, however, could violate physical laws, and its predictions could be infeasible. For instance, the input–output relationship needs to be monotonic. In this case, monotonicity constraints need to be incorporated into GPRs. Derivatives can be included as a co-kriging approach where both the values and the gradients of the target function are predicted, and the derivatives are then constrained to be only positive or negative [127,128]. Inequality and convexity constraints can also be incorporated in GPRs using truncated Gaussian distributions so that the posterior predictions are bounded [129–131]. In this case, Markov chain Monte Carlo sampling is used for Bayesian model update.

By integrating domain-specific physical knowledge, such as PDEs, analytical equations, initial and boundary conditions, into the GPR framework including kernel functions and the type of distributions, the model's adherence to fundamental physical laws can be significantly enhanced. Classical "physics-in-the-prior" approaches replace or augment the GP mean function with a mechanistic model or design new kernels that encode physical structure. For example, Hassani et al. [132] embedded a degradation law in the mean function to achieve accurate fault diagnosis and remaining-useful-life prediction for rolling-element bearings. Similarly, Long et al. [133] directly used PDEs as the mean functions to construct GPRs. Zhu et al. [134] constrained the GP mean function with a tool-wear model to improve extrapolation stability in manufacturing settings. To predict the acoustic and dynamic behaviors of structures, Jones et al. [135] designed a new GPR kernel function based on the Laplacian eigenfunctions. Li et al. [136] designed a new kernel with the additive features of hydrodynamic characteristics. Chang and Zeng [137] directly used a neural network to learn and customize the kernel function, whereas physics was embedded with the Boltzmann-Gibbs distribution instead of the Gaussian. For solid mechanics problems, Frankel et al. [138] embedded the rotational invariance of the stress-stretch constitutive relation in GPR so that the strain energy density can be predicted with fewer samples. Fuhg and Bouklas [139] generalized the tensor function representations of stresses to anisotropic materials, and a small number of inducing points were selected to construct local GPRs to improve scalability.

GPR can also be applied as a probabilistic learning approach to solve PDEs, where the likelihood that the residuals are minimized and boundary/initial conditions are satisfied can be evaluated at certain collocation points. This approach can be regarded as operator learning under probabilistic and soft constraints. For example, Raissi et al. [140] utilized a GPR model to learn the nonlinear relationship between the spatial locations and the PDE solution, as well as the load vector. Chen et al. [141] used the GPR surrogates to learn PDEs where the temporal and spatial derivatives are explicitly mapped to the ones of GPR models. Pförtner et al. [142] showed that the GPR-based operator learning can be applied and integrated with collocation, spectral, and Galerkin methods to solve linear PDEs. Hamelijnck et al. [143] constructed GPR models to predict both state variables and the derivatives in linear and nonlinear PDEs. Dalton et al. [144] further extended this approach by introducing hard constraints of domain boundaries based on R-function so that the solution domain can be further generalized to have arbitrary shapes.

4.3 Physics-Informed Generative Models. Conventional generative models such as VAE [145], generative adversary

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networks (GANs) [146], and diffusion models [147] have shown considerable promise in capturing complex data distributions and enabling tasks such as uncertainty quantification, data reconstruction, and synthetic data generation. However, these models often operate as black boxes, lacking adherence to the fundamental physical principles that govern the systems under study. This deficiency can lead to outputs that are statistically plausible yet physically inconsistent, thereby limiting their applicability in engineering and scientific problems where the laws of physics provide essential constraints.

To address this critical limitation, the concept of physics-informed generative modeling has emerged as a powerful paradigm. By embedding physical laws into the structure and training objectives of generative models, these approaches ensure that the generated outputs are not only data-consistent but also physically valid. Among these models, physics-informed variational autoencoders (PI-VAEs), physics-informed generative adversarial networks (PI-GANs), and physics-informed diffusion models (PIDMs) have attracted particular attention due to their complementary strengths and wide-ranging applicability.

4.3.1 Physics-Informed Variational Autoencoder. In PI-VAEs, prior physical knowledge is embedded directly into the latent variable structure and learning objectives. This integration provides both interpretability and regularization, ensuring that the generative process aligns with established physical principles. The general formulation of PI-VAE augments the traditional evidence lower bound (ELBO) by incorporating a physics-informed residual term as a soft constraint, yielding the training objective as

$$\mathcal{L}_{\text{PI-VAE}} = \mathbb{E}_{q_\phi(\tilde{z}|x)}[\log p_\theta(x|\tilde{z})] - D_{\text{KL}}(q_\phi(\tilde{z}|x)\|p(\tilde{z})) + \gamma\mathcal{L}_P \quad (15)$$

where x denotes the observed data as input, and $\tilde{z}$ is the latent variable. $q_\phi(\tilde{z}|x)$ represents the variational posterior parameterized by ϕ. $p_\theta(x|\tilde{z})$ is the likelihood function modeled by the decoder. The prior $p(\tilde{z})$ is typically Gaussian, while γ regulates the influence of the physics-informed residual loss $\mathcal{L}_P$, which penalizes deviations from known physical laws. This formulation ensures that the model learns latent representations that balance data reconstruction with physical consistency, providing a foundation upon which more sophisticated variants are built.

Hora et al. [148] applied the PI-VAE to reconstruct turbulent wall-bounded flow fields from sparse planar measurements. By incorporating physics-based regularization, their model accurately recovered 3D flow structures, outperforming conventional methods in preserving small-scale turbulence features and statistical accuracy. Kulkarni and Sabato [149] employed a similar PI-VAE framework for structural health monitoring, focusing on full-field thermal expansion and damage detection in metallic structures. The model, trained solely on healthy data, effectively identified defects in both simulated and experimental scenarios, demonstrating strong generalization and robustness to the sparsity of the measurements. In the field of industrial fault detection, Lai et al. [150] combined physics-based simulations with a PI-VAE to detect anomalies in electro-hydraulic servo actuators used in aircraft engines. Their method outperformed traditional autoencoders and physics-only models, achieving more reliable fault detection with reduced false alarm rates under limited data availability. Zhong and Meidani [151] extended the basic PI-VAE to solve forward and inverse problems involving stochastic differential equations (SDEs). Using maximum mean discrepancy for latent space alignment, they demonstrated the model's ability to efficiently learn complex stochastic dynamics and provide accurate uncertainty quantification.

A notable extension of the basic PI-VAE is the vector quantized PI-VAE (VQ-PI-VAE), which introduces discretization into the latent space through vector quantization. The loss function is augmented as $\mathcal{L}_{\text{VQ-PI-VAE}} = \lambda_D\mathcal{L}_D + \lambda_{\text{VQ}}\mathcal{L}_{\text{VQ}} + \lambda_P\mathcal{L}_P$. In this expression, $\mathcal{L}_{\text{D}}$ promotes fidelity to the observed data, and $\mathcal{L}_{\text{VQ}}$ enforces closeness to discrete latent codebook entries. $\mathcal{L}_P$ maintains adherence to physical constraints. The coefficients λ_D, λ_{VQ}, and λ_P balance the influence of each term. This variant has proven effective in applications such as fusion plasma simulations [152], where substantial improvements in data compression have been demonstrated while preserving the essential physical characteristics of the system.

Another variant of PI-VAE called a physics-informed dynamical variational autoencoder (Φ-DVAE) model [153] addresses the assimilation of high-dimensional and unstructured observations into dynamical systems governed by known physics, while the observation operator and physical parameters remain unknown. The model factorizes the joint distribution of observations, latent embeddings, physical states, and parameters as

$$p(y_{1:N}, x_{1:N}, u_{1:N}, \Lambda) = p(\Lambda)\prod_{n=1}^{N} p(y_n \mid x_n)\, p(x_n \mid u_n)\, p(u_n \mid u_{n-1}, \Lambda) \quad (16)$$

Here, the pseudo-observations x_n act as learned embeddings bridging the raw observations y_n and the physical latent states u_n, which evolve according to known dynamical laws parameterized by Λ. The encoder neural network maps observations to the latent space, while the latent state posterior is approximated using an extended Kalman filter. The physical parameters Λ are jointly inferred with the latent states and embeddings.

Training maximizes the ELBO, which balances the reconstruction likelihoods of observations and pseudo-observations, the physical consistency of latent dynamics, and the regularization of variational distributions. This framework is well-suited for applications such as unstructured data assimilation, physical parameter inference, and long-term forecasting in dynamical systems. It has been demonstrated on systems like Lorenz-63 and the Korteweg–de Vries equation, successfully recovering system parameters and latent dynamics from high-dimensional observations, while providing uncertainty quantification and enabling physically consistent extrapolation.

The dual latent variable formulation introduced by Takeishi and Kalousis [154] represents a significant advancement in the PI-VAE framework, offering a structured separation between physics-informed and data-driven representations within the latent space. This architecture employs two latent variables: z_P, encoding physical parameters, and z_A, capturing residual, nonphysical complexities. The priors are specified as $p(z_P) = \mathcal{N}(m_P, v_P^2 I)$ and $p(z_A) = \mathcal{N}(0, I)$. The generative process integrates both components as $x = F[f_A, f_P; z_P, z_A] + \varepsilon$. In this formulation, f_P represents the deterministic physics model. f_A is a neural network modeling unstructured component. F combines the outputs of these functions, and ε accounts for observational noise.

The encoder approximates the posterior factorization as

$$q_\psi(z_P, z_A|x) = q_\psi(z_A|x) \cdot q_\psi(z_P|x, z_A) \quad (17)$$

The training objective extends the ELBO to include separate Kullback–Leibler (KL) divergences for each latent space.

To preserve the meaningful contribution of the physics model f_P, Takeishi and Kalousis [154] introduced additional regularizers: the regularization of predictive posterior consistency and two stages of physics encoder alignment (RDA-1 and RDA-2). These regularizations ensure that the neural network component does not overshadow the physics model and that the encoder accurately infers physically consistent latent representations. The method was applied to a range of synthetic and real-world tasks, including pendulum dynamics, advection-diffusion systems, galaxy image generation, and human gait modeling. In these diverse settings, the dual latent PI-VAE demonstrated superior generalization performance, particularly in out-of-distribution scenarios. The model exhibited strong extrapolation capabilities, accurately reconstructing trajectories and physical parameters beyond the training distribution. Moreover, the ability to manipulate the physics-grounded latent variables allowed for controllable and interpretable

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generation, enabling counterfactual analysis by directly varying physical parameters such as angular velocity or diffusion coefficients. Overall, the dual latent PI-VAE effectively balances physics-based priors and data-driven learning, yielding models that are not only accurate and robust but also capable of providing physical insight into the generative process.

4.3.2 Physics-Informed Generative Adversarial Network. PI-GANs extend the conventional GAN framework by embedding physical constraints into the adversarial training process. The foundational structure comprises a generator network that synthesizes solutions based on latent inputs and a discriminator network that seeks to distinguish between generated and real data samples. Crucially, PI-GAN integrates governing physical equations directly into the training objective as soft constraints, thereby ensuring that the generated solutions not only align with observed data but also respect the underlying physical laws of the system.

The standard formulation of PI-GAN augments the Wasserstein GAN with a gradient penalty objective to promote stable training. The generator loss is given by

$$\mathcal{L}_G = -\mathbb{E}_{\tilde{z}\sim P_{\tilde{z}}}[D(G(\tilde{z}))] \tag{18}$$

where $G(\tilde{z})$ represents the generator output for a latent variable $\tilde{z}$, and $D(\cdot)$ is the discriminator output. The discriminator loss is defined as

$$\begin{aligned}\mathcal{L}_D = {} & \mathbb{E}_{\tilde{z}\sim P_{\tilde{z}}}[D(G(\tilde{z}))] - \mathbb{E}_{x\sim P\,data}[D(x)] \\ & + \lambda \mathbb{E}_{\hat{x}\sim P_{\hat{x}}}\left(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1\right)^2\end{aligned} \tag{19}$$

where λ regulates the strength of the gradient penalty, enforcing the Lipschitz constraint critical for stable GAN training. To incorporate physics, PI-GAN introduces an additional penalty term that penalizes violations of the governing physical laws, such as

$$\mathcal{L}_P = \frac{1}{N}\sum_{i=1}^{N}\left\| \mathcal{N}_x[u(x_i; G(z))] - f(x_i)\right\|^2 \tag{20}$$

where $\mathcal{N}_x$ represents the differential operator of the governing equation, and $f(x_i)$ is the known source term or forcing function at point x_i.

PI-GAN has been applied across a range of complex physical systems. Its effectiveness has been demonstrated in solving high-dimensional SDEs [155–157] and PDEs [158]. By enforcing stochastic PDE constraints within the GAN framework, the model accurately learned solution manifolds and their associated uncertainties, even in the presence of sparse or noisy data. Compared to PINNs, PI-GAN exhibited superior stability in approximating stochastic solutions due to its adversarial training mechanism.

In the domain of computational mechanics, PI-GAN has been effectively deployed for material property identification. Warner et al. [159] utilized PI-GAN to infer spatially varying elasticity fields from limited displacement observations. The integration of elasticity PDE constraints ensured that the predicted material fields were both data-consistent and physically plausible, enabling accurate inverse reconstruction of heterogeneous materials.

Further extending its application, He et al. [160] employed PI-GAN for multiaxial fatigue life prediction in engineering structures. Their framework incorporated well-established fatigue models within the GAN training objective, enabling the generator to produce life predictions that adhered to empirical fatigue criteria while capturing data-driven uncertainties. The resulting model outperformed conventional empirical approaches, offering improved accuracy in fatigue life estimation under complex loading conditions.

In fluid dynamics, PI-GAN has been adopted for multiphase [161] and turbulence super-resolution tasks. Subramaniam et al. [162] demonstrated that PI-GAN could generate high-fidelity, high-resolution turbulent flow fields from coarse input data by embedding the Navier-Stokes equations into the training objective. This approach effectively captured fine-scale turbulence structures and improved energy spectrum recovery compared to baseline super-resolution methods.

Collectively, these studies underscore the versatility and effectiveness of PI-GAN in learning physically consistent generative models across a diverse set of applications. The integration of physics constraints within the adversarial training paradigm enables PI-GAN to maintain data fidelity while ensuring compliance with governing equations, leading to robust performance even in settings characterized by data sparsity, high dimensionality, and complex physical behaviors.

4.3.3 Physics-Informed Diffusion Model. Diffusion models, rooted in a probabilistic framework, were originally developed for image generation. They have shown remarkable success in capturing complex data distributions through purely data-driven training [147]. Although this approach ensures that the generated samples are statistically consistent with the training data, it does not guarantee compliance with the underlying physical principles [163,164]. Therefore, PIDMs were developed to extend traditional denoising diffusion models by incorporating physical constraints into the training process.

The theoretical basis of the diffusion model comes from the non-equilibrium thermodynamics and the probabilistic process [147,165]. Classic diffusion models, such as denoising diffusion probabilistic models (DDPMs), can gradually transform the original data distribution into a standard normal distribution. It is achieved by constructing a forward diffusion process that gradually adds Gaussian noise to the data in the form of a Markov chain. Then, neural network models, such as U-Net, are employed to explore the hierarchical structures of noisy data and generate new clean samples that statistically resemble the original data. At their core, diffusion models operate through a forward process that gradually adds noise to data, and a reverse process that learns to denoise. The forward process progressively adds noise to the data x_0 until $x_T \sim \mathcal{N}(0, I)$. This process follows the Markov chain that transforms a data sample $x_0 \sim q(x_0)$ into a set of noised samples x_t in T steps by

$$q(x_t \mid x_{t-1}) = \mathcal{N}(x_t \mid \sqrt{\alpha_t}\, x_{t-1}, (1-\alpha_t)\mathbf{I}) \tag{21}$$

where α_t is a noise schedule controlling the gradual corruption of data, and $\mathbf{I}$ is the identity matrix ensuring isotropic noise. The noise schedule is usually predefined to ensure a smooth transition from data to pure noise over T steps. The reverse process aims to gradually remove noise from a purely noised data x_T to recover a sample that has the same distribution as x_0. This process is achieved by a conditional probability distribution $p_\theta(x_{t-1}|x_t)$, which is parameterized by neural networks. The training of the neural network is performed by optimizing the usual variational bound on negative log-likelihood. In actual training, the diffusion model usually employs this approach by

$$\mathcal{L} = \mathbb{E}_{x_0,\varepsilon_*,t}\left[\|\varepsilon_* - \varepsilon_\theta(x_t, t)\|^2\right] \tag{22}$$

where ε_* is random noise following a normal distribution $\mathcal{N}(0, \mathbf{I})$, and ε_θ is a function approximator intended to predict ε_* from x_t.

PIDMs combine the generative capabilities of diffusion models with the physical constraints of PINNs, aiming to overcome the limitations of both approaches. Bastek et al. [163] introduced a method for integrating physical laws by regularizing the loss function. Specifically, the physical constraint is expressed as a residual operator $R(x_0) = 0$, which is incorporated as an additional loss term. Thus, alongside optimizing the standard diffusion model objective $\mathbb{E}_{x_0\sim q(x_0)}[\log p_\theta(x_0)]$, a physical constraint term $\mathbb{E}_{\hat{r}}[\log p_\theta(\hat{r})]$ is also optimized, where $\hat{r}$ represents the residual of the governing equation $R(x_0) = 0$. Another prominent approach to incorporate physical laws into diffusion models is through conditional diffusion models, which was proposed by Shu et al. [166]. This method incorporates physical laws as conditional information within the DDPM framework. The idea can be expressed as

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$p_\theta(x_{0:T}) = p(x_T)\prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t, \frac{\partial r_t}{\partial x})$, where $\frac{\partial r_t}{\partial x}$ serves as the physics-informed condition. In this formulation, the guidance is provided by a learned combination of the residual gradient and noise in the DDPM. This approach has proven effective in high-fidelity computational fluid dynamics data reconstruction from low-fidelity inputs. However, these methods are typically limited to static 2D PDEs and cannot handle spatiotemporal dynamics.

To address this limitation, Pi-fusion [167] was proposed to incorporate PDE loss into the overall loss function using denoising diffusion implicit models [168]. This integration brings temporal information into the training process. Pi-fusion is capable of solving spatiotemporal Navier–Stokes equations. In contrast to Pi-fusion's approach, the projected generative diffusion models (PGDMs) [169] enforced physics-informed output generation during the sampling process via a projector-guided operation. This ensures that the generated samples satisfy a set of physical constraints. Similarly, Jacobsen et al. [164] integrated physical laws during the sampling stage and adopted a physically consistent sampling method to further minimize the PDE residual of the generated output. Table 2 provides an overview of these PIDMs.

4.4 Operator Learning and Symbolic Regression

4.4.1 Physics-Informed Operator Learning. Operator learning, such as DeepONet [170] and FNO [125], aims to learn the mapping (i.e., an operator) from one function space to another. It leverages data-driven methods, particularly neural networks, to approximate operators and address high-dimensional problems. Physics-informed operator learning combines neural operators with physical constraints to achieve better performance compared to PINNs or data-driven operator learning alone. The integration of operator learning and PINN has proven to be highly effective not only in solving forward and inverse problems of PDEs [171,172] but also in the simulations of fluid dynamics, heat transfer, and other scenarios [173–175].

One of the representative operator learning algorithms is DeepONet [170], which decomposes an operator into two components: feature extraction from the input function and basis function representation at spatial points. It employs a branch network to encode input function features and a trunk network to learn the basis functions, with the final output obtained via their inner product. This decomposition significantly reduces the complexity and computational cost of operator mapping. FNO [125] is another very effective operator learning framework. It uses the Fourier transform to map functions into the frequency domain, where operator mapping is learned. The architecture of FNO includes a lifting layer, multiple Fourier layers, and a projection layer. The lifting and projection layers are used for the linear transform, while the Fourier layers are used to extract the refined features. Moreover, there are other operator learning algorithms, such as neural operator [176], learning operators with coupled attention [177], and graph neural operator [178]. These operator learning algorithms demonstrate strong efficiency and generalizability. However, they are highly data-dependent, computationally expensive, and often lack physical consistency because of the absence of embedded physics information.

A key focus in operator learning is to embed physics so that the models remain physically consistent. Traditionally, this is achieved by augmenting the training objective with PDE residuals and boundary/initial conditions (in strong or weak form), enabling optimization in low- or no-label regimes and improving out-of-distribution extrapolation. Representative frameworks include physics-informed neural operator [175], physics-informed DeepONet [179], multifidelity and Bayesian operators [180], and finite operator learning [181]. More recently, hard enforcement of physics at the architectural/discretization level has gained traction. Structure-preserving operator networks [182] were developed to integrate operator networks within a finite element discretization to preserve key properties of the continuous system and robustly handle complex geometries and boundary conditions. Conservation-preserved FNO [183] introduced a learnable correction/projection that maps network outputs onto the solution subspace satisfying specified conservation laws, achieving exact conservation without relying on soft penalties in the loss function. Complementing these, Harandi et al. [184] proposed a label-free spectral approach that enforces mechanical equilibrium in Fourier space via a Lippmann–Schwinger formulation, reducing AD overhead and improving speed/accuracy for heterogeneous materials. Finally, Boya and Subramani [185] presented a physics-informed transformer neural operator. By stacking iterative integral-operator blocks with cross-attention, it learns condition-aware representations that generalize effectively to unseen initial/boundary conditions, achieving low errors on fluid dynamics PDE benchmarks.

4.4.2 Physics-Informed Symbolic Regression. Physics-informed symbolic regression (PISR) is an extension of conventional symbolic regression (SR), which aims to automatically derive analytical expressions from observed data that capture the relationships between input and output variables [186]. By incorporating physical prior knowledge into the symbolic regression process, PISR enhances model interpretability and ensures consistency with known physical laws [9,187]. Its primary objective is to uncover underlying physical principles in a symbolic form, even in the presence of noisy or sparse data.

Unlike traditional regression methods, such as linear regression and polynomial regression, SR does not presuppose a specific form of a model, but rather searches the space of mathematical expressions to find the best symbolic function. The output is usually an interpretable analytical expression rather than a numerical prediction or black-box mapping. Given a dataset $\{(x_i, y_i)\}_{i=1}^{N}$, SR is able to find a function $f(x)$ that achieves $f(x_i)$ to approximate y_i

Table 2 Comparison of different PIDMs based on their physics-informed strategy, training, and sampling methods

Model	Physics-informed strategy	Training	Sampling
PIDM [163]	Embedding PDE residual loss in training, and adjusting denoising direction during sampling	Incorporating PDE residual loss into total loss	Guiding denoising with physical residual (gradients of posterior distribution)
Pi-fusion [167]	Embedding PDE residual loss in training, and adjusting denoising direction during sampling	Incorporating PDE residual loss into total loss	Guiding denoising with physical residual
PIDM-Flow [166]	Embedding physics as a condition in training, and using conditional info to enhance consistency in sampling	Training with loss combining data and physics, and only optimizing data distribution	Optimizing consistency with boundary/initial conditions
PGDM [169]	No physical info in training. Using projection operations to enforce physical constraints during sampling	No physical loss, and only optimizing data distribution	Projecting samples to satisfy physical constraint set
CoCoGen [164]	No physical info in training. Using residual gradients to adjust samples during sampling.	No physical loss, and only optimizing data distribution	Adjusting generated samples by residual gradients

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as much as possible, while keeping the expression simple and generalizable at the same time. The optimization objective for SR is based on minimizing the data fitting error. The complexity penalties such as expression length or number of operators are included to avoid overfitting [188]. The expression of SR can be written as

$$f^* = \arg\min_{f \in F^*} \sum_{i=1}^{N} \left(y_i - f(x_i)\right)^2 \tag{23}$$

where x_i represents the input data. y_i is the corresponding output. $f(\cdot)$ signifies the mathematical expression to be learned from the function space F^*. f^* indicates the optimal expression that minimizes the error.

By incorporating physical knowledge into the SR process, PISR is developed to uncover underlying physical principles in a symbolic form [9,187]. Compared to traditional SR, PISR shrinks the hypothesis space and thus tends to return simpler, more explainable expressions with improved data efficiency [189]. A key approach in this area is sparse identification of nonlinear dynamics (SINDy), which fits a sparse set of terms from a dictionary to recover governing ODEs [190]. Subsequent variants have extended this concept to PDE discovery, often referred to as PDE functional identification of nonlinear dynamics or PDE-SINDy [19]. Additionally, Deep-SINDy [191] combines SINDy with learned coordinates via autoencoders, while weak-form SINDy adopts weak-form testing to enhance robustness under noisy and irregular sampling [192]. In contrast to sparsity approaches, artificial intelligence (AI)-Feynman fuses dimensional consistency, functional separability, and symmetry with a small neural network to prune the search space before symbolic solving, which lowers computational cost and improves out-of-distribution fidelity [9].

PISR has demonstrated significant advantages across various applications. A recently developed open-source computational framework called scientist-machine equation detector demonstrated a modular "scientist-in-the-loop" pipeline. It allows for components, such as prior filters, simulator checks, the SR core, and validation, to be easily adjusted. With stronger priors, it achieves high efficiency and accurately recovers physical laws from challenging sedimentation experiments that involve nonlinear drag, noise, and missing features [193]. In engineering domains, PISR has been adapted to model the tool-wear process with domain constraints [194] and analyze nonlinear wave dynamics using physics-guided bases and weak residuals [195]. For materials screening, the sure-independence screening and sparsifying operator offer a PISR-style compressed sensing framework. This framework discovers low-dimensional, physically consistent descriptors for phase discrimination and alloy design, enabling the creation of interpretable maps for rapid screening [196].

4.5 Physics-Informed Graph Neural Networks. GNNs are a class of deep learning models that naturally capture the topological or spatial relationships between elements. In various engineering applications, the elements and relationships are usually modeled as nodes and edges of graphs, respectively, such as networks, power grids, supply chains, mesh models, grain structures, etc. The learning of GNNs involves node update and message passing between nodes. Particularly, the learning of convolutional GNNs has information propagation through local neighbors in the spatial domain (e.g., GraphSAGE [197], graph attention networks [198], and heterogeneous graph neural network (HetGNN) [199]) or spectral domain after graph Fourier transform (e.g., graph convolutional network (GCN) [200] and hypergraph neural network (HGNN) [201]). Special GNNs have been developed to accelerate engineering simulations, such as solving the Poisson equation [202], performing super-resolution [203], simulating mesh deformation and adaptive meshing [204], geometry evolution [205], and multifidelity graphs [206].

Physical and domain-specific information can also be embedded in GNNs to improve their performance. Similar to PINNs, the residual in the weak formulation of PDEs can be directly utilized in the loss function of GNNs [106]. Other physical information including energy, boundary conditions, and initial conditions can also be embedded in the loss function to guide the learning process [102,107,207]. For complex problems such as simulating continuum deformable body, both spatial and temporal correlations need to be considered. Additional information can be modeled as hidden states with temporal prediction approaches such as RNNs to predict the message passing and edge update process of GNNs [208]. In combination with reinforcement learning (RL), GNNs can be applied to model the network of agents [209]. GNNs have also been utilized to predict structural failures. Failure and fragment modes can be embedded in GNNs. New active learning strategies can also be introduced in the message passing so that the probability of failures can be predicted [210,211].

4.6 Physics-Informed Reinforcement Learning. RL aims to learn the optimal policy to maximize cumulative reward through interactions between an agent and the environment. Unlike traditional supervised and unsupervised learning methods that update models based on training data, RL optimizes a policy depending on trials and errors. Specifically, the agent selects actions according to the states of the environment and its policy. These actions will affect the environment, leading to state changes and rewards. In turn, the agent updates its policy to improve future responses.

To enhance the physical plausibility, effectiveness, and learning efficiency of RL for complex problem solving in the real world, physics-informed reinforcement learning techniques have been developed. Due to the unique structure and learning strategy of RL, the methods of embedding physics into the RL architecture are slightly different from those of other ML techniques. The key elements of the RL model, including state, action, reward and penalty functions, and policy and value networks, can all be modified to incorporate physical knowledge. Thus, the following subsections will detail how the physics information is embedded in these elements.

4.6.1 State Design. The environment states in RL can be more constructive for agents by considering physical knowledge. There are generally three types of methods for physics-based state design. The first method aims to combine the states of other agents with those of the target agent using the fusion strategy inspired by prior knowledge, as illustrated in Fig. 12(*a*). The second method involves fusing additional physical information with the states of the target agent. The additional information can be either known physical knowledge or calculated states based on prior knowledge, as demonstrated in Fig. 12(*a*). The last approach is to add physical constraints to the state space for selecting the specified states, which is presented in Fig. 12(*c*). The added physical constraints can be either conservative rules or the representation network that generates states in accordance with the system characteristic.

State fusion with other agents and additional information has been applied to achieve car-following control for connected automated vehicles (CAVs) [212]. The real-time motion of each CAV was influenced by motion information from downstream vehicles, roadway geometry, and ground-truth vehicle trajectories. The multipredecessor information was weighted summed according to equilibrium and consensus concepts from the control theory. This scheme achieved great lateral control accuracy and stable oscillation damping performance of CAVs. Similarly, for aircraft conflict resolution, Zhao and Liu [213] encoded real observations, such as aircraft's position, speed, and heading angle, and imaginary data generated based on physical prior into image representations to search for the optimal policy. The imaginary data include physics modeling of the prediction and conflict detection using a safety bound model. In adaptive cruise control, Jurj et al. [214] designed an additional state called the minimum jam-avoiding distance according to the agent states and prior knowledge of avoiding collisions. With this strategy, the cruise

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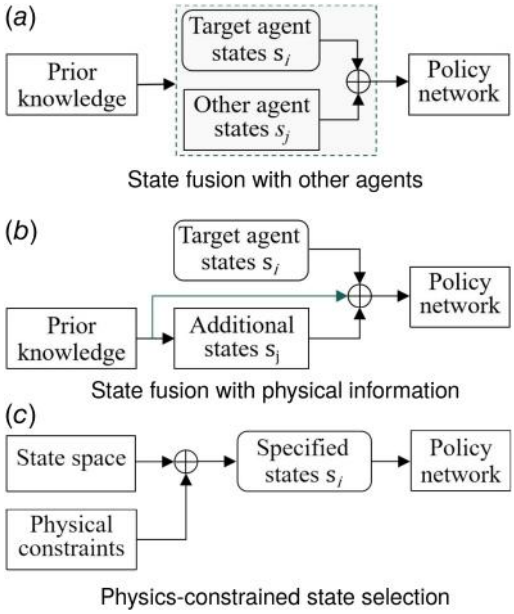

Fig. 12 Physics-based state design methods: (*a*) state fusion with other agents, (*b*) state fusion with physical information, and (*c*) physics-constrained state selection

shows fewer collisions and more equidistant travel than the results of using traditional RL.

The approach of adding state constraints has been applied in the formation control of mobile robots, where a barrier function was designed to restrict the state information that can only be perceived from neighbor agents [215]. The barrier function was incorporated into the policy, which effectively enables collision avoidance control for robots. Additionally, in voltage control of robust distribution systems, Cao et al. [216] embedded the structural correlations of measurement data by forming graph-structured data into a graph attention network to imitate the calculation process of power flow. The representation network can fill missing values or smooth noisy measurements using information from adjacent nodes and extract informative features for decision-making. Therefore, this approach improves the robustness of voltage control against anomalous measurements.

4.6.2 Action Management. The agent's actions should obey the safety protocols of the physical environment so that RL can be truly implemented in the real world. An effective method to ensure safe actions is to apply shields [217] or filters to check the rationality of actions. The shields are usually integrated with simple and conservative rules derived from practices or expert knowledge [218]. If the actions violate operation constraints, the shield can refine the actions to be appropriate. The rationality of an action can be evaluated based on the action itself or the state it produces. Additionally, physical knowledge can be reflected in the design principle of shields, safety protocols, and the process of modifying unsafe actions into safe ones.

Post hoc shielding is the most common action management method, which tries to check agent actions before applying them to the environment. Post hoc shielding has been widely used in complex operating conditions that require continuous action output, such as various energy control. For example, Chen et al. [219] developed a physics-based shielding mechanism to avoid excessive charging or discharging in safe voltage control of power distribution systems. If unsafe states occur, the shielding mechanism can replace all related actions with safe ones inferred according to the physical and electrical characteristics of the battery energy system. Meanwhile, it imposes a penalty on the agent to promote safe policies. In the cooling control of buildings, Wang et al. [218] designed a post hoc rectification, based on the thermodynamics governing the heat process, to rectify unsafe RL recommendations. An LSTM network, trained by fitting historical safe operation traces, was used to capture the thermodynamic laws and extrapolate the transitions to unsafe states. Similarly, Yu et al. [220] utilized a barrier function to adjust the power consumption of the cooling system based on the comfort constraints of the building temperature. Postposed shielding is also very critical for the control of robots and automotive vehicles. Feng et al. [221] leveraged a safety layer to achieve multi-agent collision avoidance in the robotic system. The modification operation in the safety layer is a constrained optimization problem that aims to find the action closest to the identified unsafe one while satisfying the constraint of safety distance. This strategy has also been utilized in automotive vehicle control [222,223], where the safety constraints were defined according to driving dynamics, traffic rules, and physical priors, such as speed and distance restrictions, as well as the control limit from vehicle theory.

Pre hoc shielding, also called preemptive shielding [224], is another shielding method that removes unsafe actions before training the agent. It is usually applied in scenarios that require discrete action outputs. Fisac et al. [225] developed a preemptive shielding framework by integrating Hamilton–Jacobi reachability methods from control theory with Bayesian analysis based on empirical observations. It guarantees that the search space of the learning algorithm satisfies the specified constraints by exploiting approximate knowledge of system dynamics. This method successfully enabled the safe flight of a quadrotor vehicle under strong external disturbances. Zhang et al. [226] employed preemptive shielding for energy management of hybrid electric vehicles, where a rule-based coach is designed to ensure safety training. The coach checks whether the calculated engine speed and torque can ensure that each power component in the power train satisfies the physical limits. If not, the coach generates new control signals using the rule-based energy management strategy.

Although shielding-based methods are easy to implement and have ensured the safety outputs of RL in many applications, they often lead to suboptimal policies. This is because the conservative laws incorporated may interfere with the training process [227]. One way to overcome this is to avoid the infeasibility of actions by constraining the exploration process during agent training. Inspired by this concept, Biswas et al. [228] developed a physics-informed exploration (PIE) strategy for safety energy management of hybrid electric vehicles. If infeasible actions occur, the PIE strategy chooses new actions from the predefined safety action set while updating the bounds of the safety action set according to the physical cause of infeasibility from prior experience. Otherwise, the actions will be directly calculated by a policy network embedded with prioritized experience replay (PER), which prioritizes transitions of action-state pairs from previous training. The coupling of PIE and PER improves adherence to safety constraints and reduces overall violations during training. She et al. [229] achieved safety training by mapping the output of the actor network into real-time gains using a transformation rule derived from the numerical model of the control system. This action mapping ensures that the control gains remain in a rational range defined by engineering practices, thereby helping microgrids achieve accurate trajectory tracking.

4.6.3 Reward and Penalty Design. Physical task-oriented rewards can endow interpretability to the agent's strategy [230] and help assign more accurate rewards for agent actions [231]. The related techniques usually modify reward functions according to physical objective functions or constraints to force agents to operate following specified laws. The general form of the physical reward function in most literature can be roughly expressed as (there may be slight differences in different application scenarios)

$$r_t = \zeta M(H(a_t) - H^g) + \eta w(a_t) \tag{24}$$

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where $H(a_t)$ and H^g mean the effect caused by the action a_t and the target effect at time t, respectively. $M(\cdot)$ is the function that calculates the partial reward value based on the difference between the real and target effects. $w(a_t)$ is another type of effect caused by a_t. ζ and η are the weights. Hence, this reward function guides the agent to take actions toward approaching the target effect while satisfying certain performance criteria.

The design of physical task-oriented rewards has successfully facilitated RL in tackling complex problems across various fields. For instance, in optimizing the nuclear fuel assembly, Radaideh et al. [232] shaped the reward function according to the physical rules followed by nuclear engineers. The agents are guided to position fuel rods in the assembly while adhering to constraints on the rod number and position, as well as an energy optimization strategy. This enables high-dimensional assembly while ensuring safety and economic viability. Li et al. [233] proposed a federated multi-agent RL algorithm for effective and efficient energy management of a microgrid system. The reward function was designed to achieve two physical targets: training the agent while meeting the requirements of operational cost and self-energy sufficiency. In the application of fluid fields, Dang and Ishii [231] designed a compound reward by integrating the prior theory of interfacial area (IA) transport into a Markov process to predict IA of two-phase flow. The reward function was constructed in the form of prediction error related to change trend, value precision, and inertia by integrating the physical models, experimental observations, and the inertia change of IA concentration. Yousif et al. [234] constructed a multi-agent RL model to reconstruct flow fields from noisy data, with each agent corresponding to one pixel and actions focused on selecting optimal filters. A physics-based immediate reward was formulated by combining the momentum and pressure Poisson equations with boundary conditions that represent the reconstruction error of velocity. The reward function directly mimics the denoising process of flow velocity, allowing the model to follow an optimal denoising strategy.

Physical task-oriented reward design has also been applied to promote various motion control. In the automatic operation control of the tunnel boring machine (TBM), Lin et al. [235] proposed a physical reward function derived by weighted summing the advance speed and the pressure difference between the chamber pressure and the earth pressure. This reward function allows simultaneous evaluation of both the efficiency and stability of excavation progress. A similar method has been used in Ref. [236] for the energy-efficient operation of a TBM working on soft ground. The reward function was designed to gain marks from high advance speed and lose marks from consumption of kinetic inertia energy. In a reference-guided RL technique for generating complex character animation, Peng et al. [237] defined the reward based on the characteristics of the reference motion to encourage the character to imitate known movements.

Like the design of physical task-oriented reward, penalties can also be defined in terms of physical constraints. In this way, penalties can be imposed if the exploration process violates physical constraints. This idea has been adopted by Zhang et al. [238] to optimize the charging strategy of electric vehicles. They formulated the charging scheduling problem as a constraint Markov decision process to find an optimal policy to maximize the operator's profit. The optimization process was constrained by the cumulative penalty derived by considering the limit of the power capacity and the power balance requirements of the energy system, as well as the constraints of demand satisfaction.

4.6.4 Policy and Value Function Design. An excellent policy in RL helps guide the agent in making optimal decisions according to environmental states, thereby maximizing long-term rewards and speeding up convergence. Incorporating physical knowledge of the environment into policy design is an effective way to enhance the policy, since it can ensure that the generated actions adhere to environmental constraints.

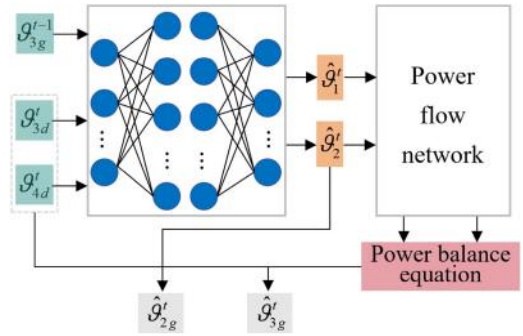

Fig. 13 Physics-informed policy

Since policy is generally a mapping network from environment states to agent actions, most existing methods for designing physics-based policy utilize techniques such as PINN, PDE, and hybrid models. These works will not be introduced in this section. A unique technique for embedding physics into the policy architecture is presented in Ref.[239], which combines FNN with a power flow network to solve the optimal power flow (OPF) problem of the renewable power system. The framework of this method is illustrated in Fig. 13. The FNN predicts voltage parameters, which are then input into the power flow network to compute operational parameters. The power flow network was built based on power balance equations to ensure that actions meet the constraints of the system. Apart from actions, the policy network generates all decision variables essential for calculating the cost function according to the inequality constraints of the operational limits. The policy network was optimized by propagating constrained gradients and integrating the policy gradient with the cost function gradient. The designed physical policy can automatically correct infeasible actions, thereby satisfying all the constraints for OPF problems.

Incorporating physical knowledge into the value function can enhance policy training and reduce computational burden. A physics-guided value function can be constructed using analytically derived physical heuristic functions. This technique has been demonstrated in Ref. [240], where the value function was designed to be a flow-control-informed function based on the self-organizing criticality formulation applied to an idealized Batchelor-Kraichnan model. The proposed scheme was applied to address the challenge of controlling a swimmer's separation from a passive target in chaotic flows, which is difficult to achieve with traditional RL methods.

4.7 Physics-Constrained Dictionary Learning. Different from general ML models that learn a surrogate model to represent the complex relationships between inputs and outputs or extract patterns from input data, dictionary learning seeks to learn an optimal dictionary that can represent dense signals by a linear combination of the dictionary and a sparse coefficient vector [241]. Incorporating physical knowledge during dictionary update allows the designed dictionary to have high representation ability even with defective input data and data acquisition restrictions in practical applications. On the other hand, considering physical constraints related to data acquisition can also facilitate the optimization of the data acquisition strategy using dictionary learning. This is because dictionary optimization algorithms endeavor to construct an optimal dictionary using the most critical and representative data points. This is valuable for improving data acquisition efficiency. For instance, when an entire temperature field needs to be reconstructed with sparse data points, prioritizing more measurements in regions exhibiting high temperature gradients enables capturing intricate temperature details. Conversely, areas with low temperature gradients necessitate fewer measurements. Moreover, when reconstructing the complete measurement field using limited

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data points, it is crucial to factor in sensor specifications such as size, resolution, sampling rate, and sensitivity for optimal results.

To integrate sensor placement strategy optimization into the monitoring of manufacturing processes, Lu and Wang [242] developed a physics-constrained dictionary learning (PCDL) method to incorporate sensor specifications as constraints when a measurement matrix Φ, a basis matrix Ψ, and a classification matrix C are optimized simultaneously. It is to solve

$$\min_{\Phi,\Psi,C} \alpha\|S-\Psi\xi\|_F^2 + \|\Phi S-\Phi\Psi\xi\|_F^2 + \beta\|L-C\xi\|_F^2 \tag{25}$$

$$\text{s.t.}\Phi = f(\Psi) \tag{26}$$

$$\|\xi_i\|_0 \le s_l \quad \forall i \tag{27}$$

$$(\Phi) \in P \tag{28}$$

where S and L represent training data and class labels, respectively. ξ represents the sparse vectors of coefficients for the original data. α and β denote the weights associated with reconstruction and classification errors, respectively. $f(\Psi)$ refers to the method, such as a constrained FrameSense algorithm [242], used to optimize the measurement matrix when the basis matrix is provided. s_l is the sparsity constraint applied to the sparse coefficient vectors. (Φ) indicates the column indices of the nonzero values in each row of the measurement matrix Φ which corresponds to the locations of the measured data points. Lastly, P is the search space, which is constrained by various sensing strategies. The framework for simultaneously optimizing the measurement, basis, and classification matrices is shown in Fig. 14. With the training data and initial basis matrix Ψ_0 as the input, the constrained FrameSense algorithm was applied to optimize the measurement matrix Φ by incorporating physical constraints related to the sensing strategies [243]. For instance, when the single-probe sensing strategy is used to collect point-wise measurements, the minimum pair-wise distance between the measurement locations is limited by a threshold value. Thus, the redundant information included in the adjacent measurements is reduced. With the optimized measurement matrix and the initial classification matrix C_0, K-means singular value decomposition (K-SVD) was performed to obtain the updated basis and classification matrices. These processes are repeated until the results are converged. With the optimized matrices, the original high-resolution data can be obtained as $\Psi\xi$, where γ is recovered by an orthogonal matching pursuit algorithm. The machine states can also be identified as $C\gamma$.

PCDL has also been applied to quantify porosity distribution in laser powder bed fusion of copper [244] and classify the machine conditions in the fused filament fabrication process measured by vibration signals [245,246]. With less redundant information in the sparse coefficient vector, both the classification accuracy and efficiency can be improved. To monitor the powder spattering in the laser powder bed fusion process with optical images, additional features such as edges in the images were used to further enhance the classification accuracy [243]. The results using the extended feature-based physics-constrained active dictionary learning (FB-PCADL) method are shown in Fig. 15. PCDL has also been adapted to combine vibration signals from multiple sensors to detect bearing faults in rotating machinery [247]. In this application, the measurement, basis, and classification matrices for each sensor were optimized individually. Subsequently, the classification matrices from all sensors were fused together, resulting in improved fault identification with higher classification accuracy compared to using a single sensor. With the optimized sampling rate, the rotating machinery condition can be monitored with much less data without sacrificing the collected information.

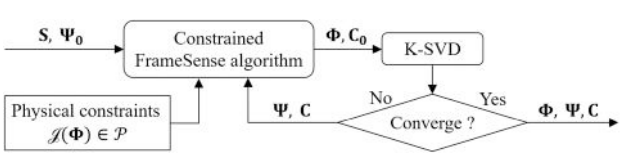

Fig. 14 PCDL framework to optimize measurement, basis, and classification matrices

While the previous discussion primarily focuses on integrating physical knowledge into the sampling strategy in PCDL, the physical knowledge can also be effectively used to construct the basis matrix. This approach can enhance the sparsity of the original data following transformation in a specific physical domain, allowing for the elimination of most redundant information during data processing. Furthermore, the basis matrix can be extended to higher dimensions to capture more correlations among high-dimensional data.

4.8 Physics-Based Compressive Sensing. While many PIML methods primarily emphasize the forward prediction of the complete physical field with simulated or experimental data, the integration of physical insights into data-driven models also enables the use of measured data to extrapolate unobservable information.

Compressed sensing is known as a pure data-driven approach to improve data acquisition efficiency by taking advantage of the sparsity representation of signals after predefined transformations such as the Fourier transformation and the wavelet transformation. To further enhance the compression ratio and reconstruction accuracy of traditional compressed sensing, Lu and Wang developed a physics-based compressive sensing (PBCS) method to reconstruct the complete measurement field based on limited measurements by incorporating the knowledge of physical phenomena in specific applications [248,249]. The general procedures of PBCS consist of two main steps, as shown in Fig. 16. In the recovery step, an inverse problem is addressed to recover the load vector ξ within the discretized physical model, utilizing sensor measurements y^* typically gathered at the boundaries of the domain. The reconstruction step then involves solving the forward model to predict the complete distribution of the physical quantity as the latent field. This approach allows PBCS to seamlessly integrate modeling in the digital space with sensing in the physical space. Essentially, the inverse problem in the recovery step in PBCS is to solve

$$\min \|y^* - \Phi y\|_2 \tag{29}$$

$$\text{s.t. } y = f(t, \xi, d, \dot{x}, \triangle d, \dots) \tag{30}$$

$$\|\xi\|_0 < s_l \tag{31}$$

where $f(t, \xi, d, \dot{x}, \triangle d, \dots)$ describes the relationships among physical quantities d as well as their time and spatial derivatives $(d, \dot{x}, \triangle d, \dots)$. Typically, this function represents physical models, which are often formulated as PDEs. The minimization process can be based on the criteria of l_2 norm. There are numerous algorithms available for recovering the load vector ξ. These include basis pursuit [250], least absolute shrinkage and selection operator (LASSO) [251], least angle regressions (LARS) [252], nuclear norm minimization [253], matching pursuit [254], orthogonal matching pursuit [255], CoSaMP [256], Fourier sampling algorithm [257], focal underdetermined system solver (FOCUSS) [258], and others.

PBCS has been applied to reconstruct the 3D temperature field of the printed part in the fused filament fabrication process. With a small set of temperature measurements on the printed part's surface, the boundary conditions, including heat flux, convection, and radiation, represented as a condensed load vector of the heat transfer model, were recovered with a high compression ratio compared to the traditional compressed sensing. After that, the complete 3D temperature field can be reconstructed by the forward modeling. Due to enhanced data acquisition efficiency, PBCS

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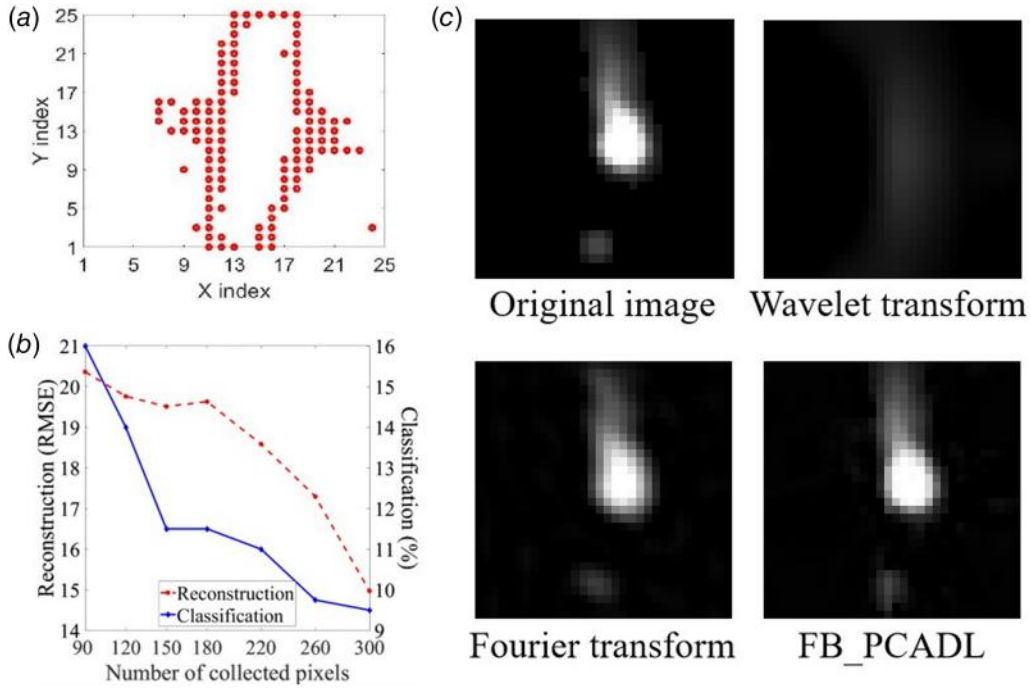

Fig. 15 (*a*) Optimized sampling locations by FB-PCADL, (*b*) classification accuracy of powder spattering in selective laser melting, and (*c*) comparison of reconstruction performance of FB-PCADL with other existing methods [243]. Reprinted with permission from Lu et al. © 2023 Elsevier.

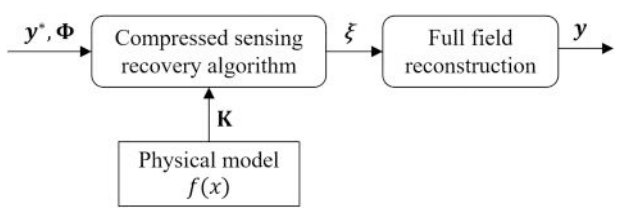

Fig. 16 PBCS framework to recover the load vector and reconstruct the complete physical quantities

has facilitated the development of digital twins for monitoring the fused filament fabrication process, enabling seamless integration between physical experimentation and digital modeling [259]. The results in monitoring the real-time temperature field in fused filament fabrication are shown in Fig. 17. A similar concept has been extended to reconstruct the complete velocity field within a wind tunnel by capturing a sparse set of point-wise flow velocities near the tunnel boundaries [260]. The detailed recirculating flow over a backward-facing step was also captured. PBCS has also been applied to monitor the melt pool within the laser powder bed fusion process by incorporating a multiphysics model for the thermal-fluid field. PBCS demonstrated the capability to reconstruct both the temperature and fluid fields of the melt pool solely through surface temperature measurements [261]. In addition to process monitoring, PBCS has also been extended to inversely identify internal heat sources based on the surface temperature measurements [262,263].

PBCS can be regarded as a sensing algorithm well-suited for handling real-time and streaming data, as it operates without the need for model training. This characteristic makes PBCS a

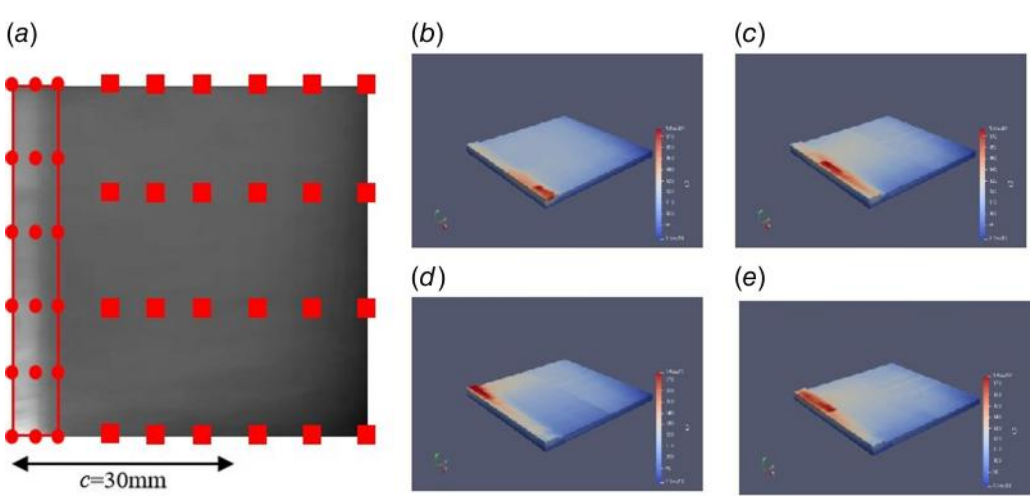

Fig. 17 (*a*) Sampling strategy of PBCS to reconstruct real-time 3D temperature field in fused filament fabrication; (*b*)–(*e*) reconstructed temperature fields at 1/9 s, 5/9 s, 1 s, and 13/9 s, respectively [259]

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highly robust approach for enhancing sensing efficiency, allowing for the accurate reconstruction of complete physical fields using only a limited number of measurements and the reliable prediction of hidden or unmeasured physical quantities. In addition to the applications mentioned above, PBCS also has significant potential in various scenarios. For instance, PBCS can be applied to detect internal cracks or stress concentrations in bridges, buildings, or aircraft using limited surface strain or vibration data. In reconstructing pollution or temperature fields in the atmosphere, PBCS can be integrated with sparse sensor networks for real-time environmental assessment. In battery or electrochemical systems, the internal state of batteries can be reconstructed through limited surface measurements for predictive maintenance.

4.9 Other Generic Techniques for Neural Network. Apart from the unique techniques of embedding physics into specific ML models described in the previous sections, there are also some generic techniques suitable for all neural network models. These techniques try to assign physical meaning to the parameters, components, and neurons of neural networks, which will be detailed in the following subsections.

4.9.1 Physics-Guided Parameter Initialization. Many neural network models need to initialize parameters before training, such as weights, biases, and learning rates. Most existing techniques choose to randomly initialize these parameters, which may result in poor convergence and getting stuck in local optima of the model. Physical knowledge-guided parameter initialization is a promising way to solve this problem, as the physics-specified parameters are more closely aligned with the underlying structure of the model and data [264].

A common approach for physics-guided parameter initialization is to use transfer learning. Specifically, the ML model is first pretrained on a related task prior to obtain the suitable parameters, which are then transferred to construct the target task model. Many works have shown to pretrain neural network models using synthetic data from simulation models. Aliakbar et al. [265] utilized this technique to initialize PINN for solving multiphysics PDEs, where the low-fidelity data from the simulation of computational fluid dynamics were used for pretraining. This approach helps to obtain reliable results without high-fidelity training data. Similarly, with the two-step multifidelity training, Tang et al. [266] successfully predicted the time-domain response of optical resonances and inferred the corresponding resonance frequencies in optical science. Read et al. [267] initialized the LSTM network to predict lake water temperatures using predictions from a 1D lake hydrodynamic model [268], which tracks state variables with Lagrangian layers. They also demonstrated that simulation-based neural network pretraining adapts better to unseen scenarios compared to purely simulation models. Park et al. [269] reported to incorporate the adjoint gradient into RL by initialization for designing highly complex optical devices. They pretrained the agent network using the figure-of-merit gradient estimated from two electromagnetic simulations, which significantly enhanced the sampling efficiency for structure optimization. Rodwell and Tallapragada [270] utilized a pretraining method to overcome the difficulty of controlling fish-like swimming robots with complex unmodeled physics. They first trained the agent network to perform increasingly complex tasks in a representative nonholonomic system to capture some primary physics. Then, the agent network was transferred to a small amount of high-fidelity data from the fluid-robot interaction system.

Using simulation models for parameter initialization is also an effective way to address the difficulties of neural networks in acquiring large amounts of costly data. However, effective initialization depends on a high-fidelity simulation model, which increases demands on both modeling complexity and computational costs. Another parameter initialization method for neural networks is to leverage numerical solutions of system-related process equations or transforms. For example, Tarbiyat and Nemati Saray [271] introduced to initialize the weights of PINN with approximate solutions of PDEs. They discretized the related differential operators and obtained an approximate solution using FDM. The initialized weights reduced the training loss of the PINN and improved the accuracy of PDE solving. He et al. [272] devised a weight initialization method based on physics-informed wavelet transform in fault diagnosis of rolling bearings. They modified the wavelet basis functions by introducing a scale smoothing factor to form a signal processing-consistent wavelet dictionary, which was then applied to initialize the weights of the first convolutional layer. This technique improved the extrapolation ability of the deep neural network model. Wang et al. [273] utilized the least squares solution, which fits the initial condition, to initialize the last layer of the physics-informed residual adaptive network (PirateNet). The adaptive residual connection in PirateNet allows the output of the network to be a linear combination of the first layer embeddings at the initial stage. Therefore, the initialized network can be progressively deepened during training. This initialization strategy enabled the encoding of appropriate inductive biases corresponding to a given PDE system within the network architecture.

4.9.2 Physical Intermediate Variable. One of the reasons for lacking interpretability in standard neural networks is that they can only extract abstract hidden variables. By introducing physical intermediate variables, certain neurons can be assigned a physical meaning, which helps to extract physically relevant representations. This technique has been applied in the voltage control of the active distribution network [274]. The researchers set the voltage out of control ratio as a target estimation variable in LSTM by adding an auxiliary-task head with the form of self-supervised learning. With this physical indicator, the hidden layers needed for training are dramatically reduced.

Another related technique is to assign some weights in neural networks with physically meaningful values and keep them unchanged during training. This approach has been applied to develop a theory-guided RNN for the modeling of seismic waveform inversion [275]. Most parameters in the RNN were forced to mimic the wave propagation process and fixed with values obtained from governing equations. The only trainable parameter is the wave velocity, which is the induced intermediate variable related to the process. The researchers demonstrated that this technique yields more robust training compared to simply introducing intermediate variables.

4.9.3 Physics-Informed Component. Improving the interpretability of neural network models can also be achieved by modifying some components in the architecture to enable them to achieve physics-related tasks. One typical example is to replace the convolutional kernels of CNN with trainable wavelets [276], which facilitates the extraction of frequency information from the input time series. Similarly, Hao et al. [277] employed a temporal causal convolution layer as the input layer of the transformer, which helps the model learn the local context of the input data. Additionally, the mathematical equation representing cutting physical knowledge was embedded in the output layer as a hard constraint. Experiments showed that the proposed method significantly improved prediction accuracy and the robustness of the transformer. In robotic control, Bahl et al. [278] employed the dynamical structure of the system, which is described by the second-order dynamic movement primitives equation of trajectory distribution, as a layer of the deep policy network. The parameters of the dynamical system were predicted as outputs of the preceding layers. The embedded structure enables the agent to explore in the space of physically plausible trajectory distributions.

In addition to layers, the activation functions in neural networks can also be assigned with a physical meaning. Howland and Dabiri [279] formulaically modified the sigmoid activation function to fit a thrust coefficient curve in the modeling of wind farm power

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output. The modified activation function can predict downstream power from upstream power, facilitating the capture of nonlinear fluidic wake interactions between successive layers of turbines. Wang and Huang [280] applied Softmax and absolute activation functions to the output layer of the actor network in RL, which maps the output coefficients to ensure compliance with stringent constraints derived from the Minkowski–Weyl theorem. This adaptation helps prevent noncompliant evaluations in the kinematics optimization of a self-propelled articulated swimmer.

5 Challenges and Opportunities of PIML

It is seen that a diverse range of PIML techniques emerged in recent years, particularly various approaches for physics embedding, knowledge embodiment and representation, and information integration. The advancement of PIML has resulted in some new approaches that can be potentially applied to design and manufacturing. However, several open issues remain unsolved, which pose critical obstacles to apply the PIML techniques to solve industry-scale problems. This section summarizes the challenges that PIML faces in the aspects of real-world applications and highlights the potential opportunities to pave the way for future research.

5.1 Challenges. Despite the advancement of PIML research in the past three decades, several significant challenges still hinder its adoption in industry-scale applications. These challenges can be categorized into several aspects, including training convergence, representation and integration of complex physical models, data imperfection, model generalizability, computational efficiency, model interpretability and reliability, and physics-data conflict.

5.1.1 Training Convergence. One of the primary challenges in PIML is how to achieve effective training convergence. Integrating physical knowledge into ML models enhances the interpretability and facilitates guided learning, which can potentially improve prediction performance. However, this incorporation of physical knowledge also introduces additional complexities and uncertainties that affect the convergence performance of model training, particularly in multiphysics and high-dimensional scenarios. The physical models that need to be tailored for specific applications can be quite complex. For instance, monitoring the melt pool states in the laser powder bed fusion process requires consideration of factors such as melt flow, temperature fields, recoil pressure, and surface tension [261]. Training a model that captures both the statistical correlations in the data and the underlying physical principles requires a significant number of iterations, and the results can easily diverge. Additionally, using a PIML model to predict physical quantities in complex scenarios necessitates a considerable number of hyperparameters, which further increases computational challenges.

Another major challenge affecting convergence is the uncertainty in the data and physical models. When the input data for the PIML model comes from experimental measurements, uncertainties such as calibration errors, instrument bias, and environmental factors are common. In such cases, the measured data may not adhere precisely to the underlying physical principles. Consequently, this can hinder the convergence of the PIML model. On the other hand, uncertainty in physical models is also inevitable, especially when the understanding of the true underlying physics is limited. All models involve simplification and abstraction, and therefore do not reflect the true physical phenomena either. Thus, the mismatch between physical models and data increases the difficulty of training.

5.1.2 Representation of Complex Physical Phenomena With Incomplete Knowledge. The representation and integration of complex physical phenomena also pose significant obstacles for PIML in real-world applications. Most of the existing PIML research efforts focus only on single-physics models. Single-physics modeling relies on assumptions and boundary conditions so that the intricately coupled multiphysics systems in real-world scenarios are simplified and the physical quantities are assumed to be independent from each other. Nevertheless, comprehensive multiscale multiphysics modeling is still necessary if we would like to obtain an accurate and detailed description of physics in modern engineering products or complex processes in advanced manufacturing. For example, the metal additive manufacturing process involves metal melting and solidification, fluid flow, heat conduction, and oxidation processes [281]. When constructing the process–structure–property relationship of metal additive manufacturing with PIML models, thermomechanical behavior, fluid dynamics, and phase change phenomena should be comprehensively considered. Relying only on simplified and approximated single-physics models and isolated knowledge may result in incoherent predictions of physical processes, thus failing to fully leverage the advantages of integrating physical knowledge in ML to effectively solve real-world problems. Constructing multiphysics models that accurately simulate complex physical phenomena is already very challenging, which requires researchers to have comprehensive knowledge and sufficient experience. Further integration with ML as in PIML undoubtedly poses great challenges.

Even with detailed dependency relationships introduced in multiphysics and nonlinear models, human knowledge about complex phenomena is still incomplete. For example, turbulent flow is characterized by high instability, featuring vorticity, diffusion, and energy dissipation processes. However, existing turbulence models still cannot capture the turbulence in atomic superfluids. There are still no comprehensive theories that consistently describe the complex phenomena [282]. Integrating a physics-based model in PIML to describe the superfluid turbulence will introduce errors and discrepancies. Thus, physics-based modeling can still face the knowledge gap and limitations. The gap and limit inevitably lead to uncertainty, which will be inherited by PIML models.

5.1.3 Data Imperfection. Similar to traditional ML, data are essential for PIML models to learn complex relationships and explore patterns of the system. The quality of data directly influences the reliability of PIML predictions as well as the training efficiency. Data scarcity is a common problem in engineering applications of ML. Data scarcity is caused by several factors, including the system complexity which requires a high-fidelity description, thus more data to capture the nonlinear input–output relationships, high costs associated with data acquisition, especially in application areas where high-throughput experimental data collection methods are not available, as well as the situation that data availability is restricted due to proprietary concerns or regulatory constraints. Because of the insufficient training data for PIML to fully explore and capture complex input–output relationships, data scarcity may also hinder the generalization of PIML to unseen scenarios [283,284].

Data scarcity is also related to data imbalance caused by the scarcity of certain classes of data. In applications such as fault and anomaly detection in manufacturing, certain states or phenomena of interest occur infrequently compared to others. For example, defects of debris, soot, and misprint in additively manufactured products rarely appear compared to normal regions [285]. Acquiring data under those rare or extreme conditions is also costly. For example, expensive computed tomography tools are necessary to detect microscopic pores in metal parts. To train ML models for classification problems, balanced amounts of training data for different classes are important for classification accuracy. Data imbalance may affect the prediction accuracy of PIML models for classification [18].

Another aspect of the data imperfection problem is the poor quality of training data. Data collected from experiments can be noisy, redundant, and incomplete, caused by systematic and random errors of the instrument, environmental disturbances,

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equipment failures, or running out of time or resources. Low-quality data not only provide limited useful information for PIML models but also distort the learning process [156]. Noisy training data can force models to replicate noise, instead of real input–output relations. If the distributions of noise differ between the training dataset and the validation dataset, the validation loss can increase. Furthermore, incomplete data may lose the key features of complete data, which can make the results of PIML models difficult to understand and interpret.

5.1.4 Generalizability. The generalization ability of PIML models is essential to their application to real-world problems. However, PIML models often suffer from limited generalizability. When geometric variation, changes of input parameters, and diversity of physical systems occur, a PIML model that is trained with data from one scenario cannot be easily reused for another.

First, when PIML is trained on a fixed geometry, it may not generalize well to new domains with different shapes or topologies. This is due to the fact that PDE solutions are highly sensitive to the geometry of the solution domain and the boundary conditions [286]. Some approaches have been proposed to address this problem, such as solution domain decomposition [287] and neural operators [125]. These methods can learn geometry-invariant representations in theory. However, in practice, the accuracy of approximations in these approaches still depends on the levels of nonlinearity or discontinuity in different application scenarios. Secondly, most of the PIML models rely on fixed input parameters such as material properties, source terms, or boundary values. However, in many engineering and scientific applications, input parameters vary significantly across different problem instances. The ability of a model to generalize across such variations is critical for its practical applicability. Methods such as parametric PINNs [288] incorporate parameters directly into the input space, enabling the model to learn a family of solutions. Moreover, operator-learning-based approaches, such as DeepONets and FNOs, are specifically designed to learn mappings from input functions or parameter fields to the distribution of solutions. However, they still cannot bypass the limitation of extrapolation inherent in ML. Finally, the PIML models are usually designed for a specific type of PDEs, ODEs, or algebraic equations. The diversity of physics exists in nature, and the physical models have a wide variety of forms, which differ in their mathematical structures. Moreover, different boundary and initial conditions can also drastically alter the solution space. Addressing this problem is challenging. Incorporating the flexible inductive biases or learning the symbolic representations of models can be helpful. However, it is extremely challenging, if not impossible, that a generic and universal PIML model is constructed by learning the physics of different domains, under the limits of model complexity and available training data, and at the same time can be specialized in solving domain-specific problems with very accurate predictions.

5.1.5 Computational Efficiency. Adding physical knowledge to ML models will increase the computational complexity of the training process, since it is similar to adding constraints in optimization or adding more criteria and restrictions in reasoning and decision-making. PIML models must strike a balance between fitting the training data and staying adherent to physical laws, which is particularly evident in the training of PINN models [289]. For PIML models other than PINNs, even though physical knowledge is not incorporated into ML through a physical loss, it can still expand and increase the parameter space that needs to be searched through during learning. These additional parameters correspond to the physical properties. When physical principles such as stress-strain relationships, dynamic stability criteria, or phase transition of materials are introduced in PIML models, they have influences on the model training so that the predictions are tailored toward the specific physical properties. This expansion of the parameter space complicates the training process and increases the required number of iterations.

Furthermore, when physical models are embedded into neural networks, the computational complexity of AD to compute derivatives increases. This process can generate additional large tensors during backpropagation, which can impose a significant burden on the transmission and storage of these tensors. Moreover, AD-based back-propagation involves long computational chains in which errors accumulate over multiple layers. The additional gradients of physical models may increase the accumulated error and impair training stability for fast convergence. Some researchers developed numerical approaches for discretization such as finite difference, finite element, and finite volume methods to calculate the embedded physical models without AD involved. However, the discretization approaches also require computational resources for mesh generation, refinement, and other additional procedures.

Computational efficiency issues are also prevalent when simulations are used to generate synthetic data for training or as inputs in the hybrid models. Especially when high-fidelity simulations are required to ensure the reliability of the synthetic data, such as finer meshes needed to discretize the computational domain, they inevitably lead to high computational costs. Parallel computation can potentially be used to speed up the training of some PIML models. Nevertheless, the added physical constraints may complicate this process. This is because the parallelization may sacrifice the coherence of physical laws in a distributed computing environment. Therefore, the computational efficiency issue must be considered when applying PIML techniques. The issue cannot simply be assumed avoidable using more computational resources. Researchers need to holistically consider how to reduce the model complexity and quantify uncertainty when incorporating physical knowledge into ML.

5.1.6 Interpretability and Extrapolation Capability. Interpretability refers to the ability of a model to provide transparent and explainable insights about the model's construction, its predictions, or causal relationship revelation. The interpretability of ML models is crucial for users to assess the reliability and suitability of the results obtained for specific applications [290]. Some methods and tools have been developed to interpret the predictions of ML models, such as local interpretable model-agnostic explanation (LIME) [291], Shapley additive explanations [292], and Captum [293]. Yet, they can only achieve the basic level of sensitivity analysis for different inputs or factors. Despite the original intention of embedding physics in PIML models is to provide explainable predictions, the interpretability of most existing PIML models is still very limited. This is because the current techniques of embedding domain knowledge in ML models are still largely based on the original black-box ML approaches, such as modifying the loss function or employing hybrid models. There are no modifications to the fundamental architecture of the ML models. As a result, the domain-specific physics information can easily get diluted and lost within the ML architecture during information propagation. Furthermore, the complexity of advanced architectures for deep learning makes it difficult to explain the process of how the ML structures and internal components were selected. Deep learning models are inherently difficult to interpret.

Another reason for the interpretability issue is that the current research priority for PIML has been on the incorporation of physical and domain knowledge to improve the prediction accuracy of pure data-driven models, instead of the improved model construction process to avoid the loss of domain knowledge in the ML models. The focus of the latter is important to enhance the interpretability of PIML models. This may require the modification of ML architectures for reliable and tractable representations of physical knowledge and more transparent data-physics integration.

It had been hoped that embedding physical and domain knowledge in ML models could improve not only interpretability, but also extrapolation capability. Yet, PIML models still suffer from the same extrapolation issue as the data-driven ML models. Early studies [11] have shown that this challenge is particularly

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predominant in the PINN models. The root causes of the extrapolation issue in PIML models are the techniques of embedding physics, similar to the interpretability issue. The mixture of losses affects the training convergence, which can lead to the violation of physical laws. Although operator learning is intended to learn general functional mappings, the reliance on the classical ML architectures makes those models inherit the undesirable ML limitations including extrapolation, in spite of some recent research efforts [294–297] that aim to enhance the temporal extrapolation capability of neural operators.

5.1.7 Physics-Data Conflicts. The data collected from experiments and simulations may contradict established physical models or principles. Physical models are constructed with necessary approximations and simplifications, whereas experimental techniques contain systematic and random errors. It is not uncommon that discrepancy exists between data and models. When training PIML models to achieve conflicting goals, it is difficult to make the models converge toward any of them. This will lead to poor convergence and inaccurate predictions. In PINNs, the convergence speeds for the reduction of losses are also sensitive to the relative weights assigned to different loss terms.

In addition, data collected from experiments may contain physical quantities of interest that are different from the ones that are directly included in the known physical laws or models. The integration of them in one PIML model would require us to establish the relation between them. When conversions between physical quantities are complex, the mathematical relations between them are often developed empirically. For instance, the density of point defects in metallic materials due to radiation can only be observed indirectly, usually by measuring the electrical resistance changes and inferring through some empirically derived relationship. In those cases, the integration of physical models and experimental measurements in PIML models can be challenging.

5.2 Applications in Industry. Although researchers have applied PIML techniques to solve engineering problems, their adoption in industry remains limited. Some attempts and case studies were made to apply PIML for problems in the manufacturing industry. For instance, in automotive production, physical-loss-based PINN models can be applied to predict thermal power in air handling units for topcoat painting and to forecast temperature profiles in degreasing tanks [298], and predict surface roughness in milling operations [299]. PINNs also have the potential to predict vibration and wave propagation in composite materials for the aerospace industry [300]. Furthermore, the hybrid model technique has been applied to predict the product completion time in production systems [301]. The physics-embedded architecture technique has been utilized to diagnose the health of aerospace structures [302], and to improve the metal removal rate in electrical discharge machining [303].

Solving industry-scale problems requires reliable, robust, and scalable PIML tools. Complex systems in industry involve interacting subsystems. Phenomena and factors at multiple length and time scales need to be considered. For example, in the aerospace industry, the effects of fluid-structure interactions, thermomechanical behaviors, structural fatigue, and corrosion need to be considered simultaneously in designing aircraft. The structural reliability, stability, and maneuverability of the aircraft need to be optimized together. To accurately describe the involved physics, engineers are required to have an interdisciplinary and deep understanding of the problems when building models for multiphysics multiscale simulations. The construction and execution of such complex physics-based models are time- and resource-consuming. Although PIML models offer cost-effective alternatives, the challenges discussed in Sec. 5.1 have been the major barriers to make the PIML a practical tool for industry applications. In addition, industry-scale models often operate in a dynamic environment. Assumptions made in academic settings cannot always be satisfied. For instance, when ML models are trained on a fixed dataset before deployment, it is assumed that the operating conditions do not change. Yet, models that present the best prediction performance only for a specified operating condition will not be useful for industry applications. As a result, it is challenging to develop a PIML model that is sophisticated enough to adapt to the diverse conditions and ever-changing processes in industries.

Therefore, the immaturity of the PIML techniques limits their applications in industry. At the same time, industry practitioners may be hesitant to adopt these new technologies due to the concerns about not only reliability, interpretability, and robustness, but also the potential disruption of the established business processes.

5.3 Opportunities. Although PIML raised significant interest in engineering research communities, the fundamental challenge of its prediction errors for large-scale problems remains. Its applicability in real-world engineering problems is still unknown. Nevertheless, this fundamental challenge offers good research opportunities for researchers at the same time. Several areas of research efforts are needed.

First, the training and learning algorithms need to be significantly improved for better convergence. The high dimensionality of the parameter space is the root cause. Efficient global optimization algorithms will allow the training to converge to global minima of losses. The complexity of training is particularly prominent for large-scale problems where multiphysics models are involved. To minimize several loss terms simultaneously, especially when these losses are intrinsically coupled and correlated, is a daunting task. There is a large gap between the needs in real-world applications and the current available global optimization algorithms. Novel methods and approaches for large-scale multi-objective optimization will alleviate the bottleneck in PIML. Furthermore, novel approaches for PIML with hard constraints are needed to tackle the challenges of training convergence, multiphysics, and extrapolation capability.

Second, the quality and quantity of the training dataset affect the training results. Research issues of imbalanced datasets for classification problems, as well as missing data, noisy data, and bias in general affect the model predictions. New approaches to mitigate the negative effects of these issues. For instance, resampling and generative approaches can be applied to alleviate the effects of these training data issues. More cost-effective utilization of training data, such as multifidelity ML models, also helps construct more reliable PIML models. In addition, active learning that cost-effectively acquires new data points for training and model updates is necessary for real-world applications. The model can be trained with the representative set of scenarios and updated when or where the physics of the problem exhibits significant variation. An effective sampling strategy can help mitigate issues such as gradient imbalance and uneven distribution of training data, which often hinder model convergence and accuracy. By prioritizing samples that capture critical features of the underlying physics, such as areas of high gradient, the model can learn more effectively and make more accurate predictions.

Third, quantifying the uncertainty in PIML models is an important research area that can improve the accuracy and trustworthiness of predictions. The majority of uncertainty quantification methods [304] are data-driven by nature. Many are also regarded as statistical ML methods, such as Bayesian inference in Bayesian networks, Markov chains in diffusion models, stochastic differential equations for generative models, Gaussian process in active learning, dimensionality reduction techniques in principal component analysis and manifold learning, and compressive sensing in dictionary learning. Uncertainty in PIML can be introduced from either data-driven or physics-based side of modeling. Model-form uncertainty and parameter uncertainty in PIML need to be differentiated so that they can be managed accordingly. Incorporating surrogate-based models such as Bayesian neural networks and

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Gaussian process can enhance the prediction capabilities of PIML and alleviate the physics-data conflicts. Dimensionality reduction can help tackle the challenge of computational efficiency in training.

Fourth, reduced-order modeling can improve the performance of PIML models. One of the ultimate goals for PIML is to speed up the predictions in comparison with traditional physics-based modeling for large-scale problems. The complexities of PIML models for these problems will be high, as complex multiscale multiphysics models are involved. Thus, methodologies for reduced-order modeling are helpful to reduce the computational complexities. The integration of ML into reduced-order modeling can achieve an acceleration of orders of magnitude faster in high-fidelity simulation [305]. Similar strategies can be taken to construct PIML models.

Finally, more effective mathematical representation methods can be developed for PIML models. Most physical models in the existing PIML techniques are static and single-physics focused, with relatively simple structures. Researchers often prefer utilizing well-established and genetic representations to creating tailored models for specific problems. Enhancing the representations to capture detailed physical processes can not only unlock the full potential of PIML but also enable the integration of multiscale and multiphysics factors. Adopting a multiscale modeling strategy that encompasses multiple time and length scales can improve both efficiency and utility of PIML models. For instance, graph theory can be used to capture the correlations between atoms in molecules, between defects in material models, between nodes in continuum mechanics models, or between subsystems in complex systems of systems. Drift-diffusion models capture the dynamics of temperature, fluids, defect densities, energy transport, spiral waves and vortices, and statistical descriptors. Furthermore, hybrid approaches [306] that integrate the learning at symbolic and syntax levels with the one at the instance level with numerical values can potentially enhance the capabilities of PIML to address some of the aforementioned challenges. The generalizability, adaptability, and flexibility of PIML models can be significantly improved.

6 Conclusion

PIML or scientific machine learning is a new research field that aims to integrate physical and domain knowledge into traditional "black-box" ML models so that the reliability, interpretability, and applicability of ML models for engineering and scientific applications can be improved. The incorporation of physics-based prior knowledge into data-driven ML models is to create more accurate and consistent representations of physical systems, thereby enhancing the training efficiency and robustness of predictions. These hybrid modeling approaches provide a promising avenue for tackling intractable problems in engineering such as design and manufacturing. This article provides a review of the current state of PIML techniques, with a focus on the methodologies of physical knowledge integration and the remaining technical challenges. The knowledge integration methodologies are stratified into three categories: hybrid models, physical-loss-based models, and physics-embedded architectures. Despite the development in the last three decades, PIML still faces major challenges of accuracy and reliability that prevent its adoption by industry. The main conclusions drawn from this review are summarized as follows.

Hybrid model techniques combine the predictions of ML and physical models in parallel or serial, with the aim of leveraging the strengths of both to improve the final prediction accuracy. Physics-based feature engineering is the most representative technique in this category, which generates physical features based on analytical models, numerical simulations, as well as feature processing and extraction approaches. Physics-based feature engineering is an effective way to overcome the difficulty of acquiring large amounts of data required by ML models.

Physical-loss-based models aim to construct a composite loss function of the neural network through automatic differentiation, numerical discretization of PDEs, and adding regularization terms. To solve computation and accuracy problems induced by the physical loss, some advanced techniques, such as a weighting strategy, a gradient improvement method, and a physics-constrained Bayesian neural network, have been developed. Physics-based loss function contributes to guiding the model training process toward adherence to physical laws.

Physics-embedded architecture is the primary technique for improving the transparency of ML, which fuses physical knowledge into the main structures of ML models. Different ML architectures have been employed in the physics embedding techniques. The typical techniques include the design of hybrid ML structures, applying physical constraints in training components, and incorporating functional components and physical variables. The essence of this category is to assign physical meanings to the individual components of ML models.

PIML still faces several major challenges in solving real-world problems, including the difficulties of training convergence and computational efficiency, how to represent and integrate physical knowledge of multiple domains, how to achieve higher transparency in model construction, and how to enhance interpretability and extrapolation capability. These challenges may inspire and drive us into new research directions to further advance PIML techniques.

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No data, models, or code were generated or used for this article.

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